

# Bayesian Machine Learning

## Course Introduction

Derek Aguiar  
University of Connecticut  
CSE5825

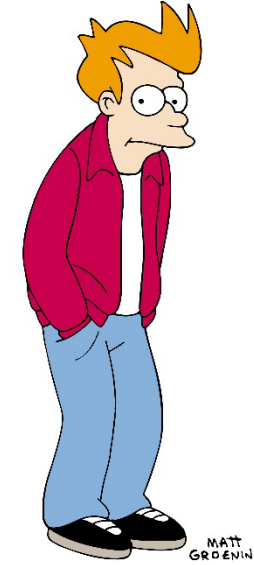
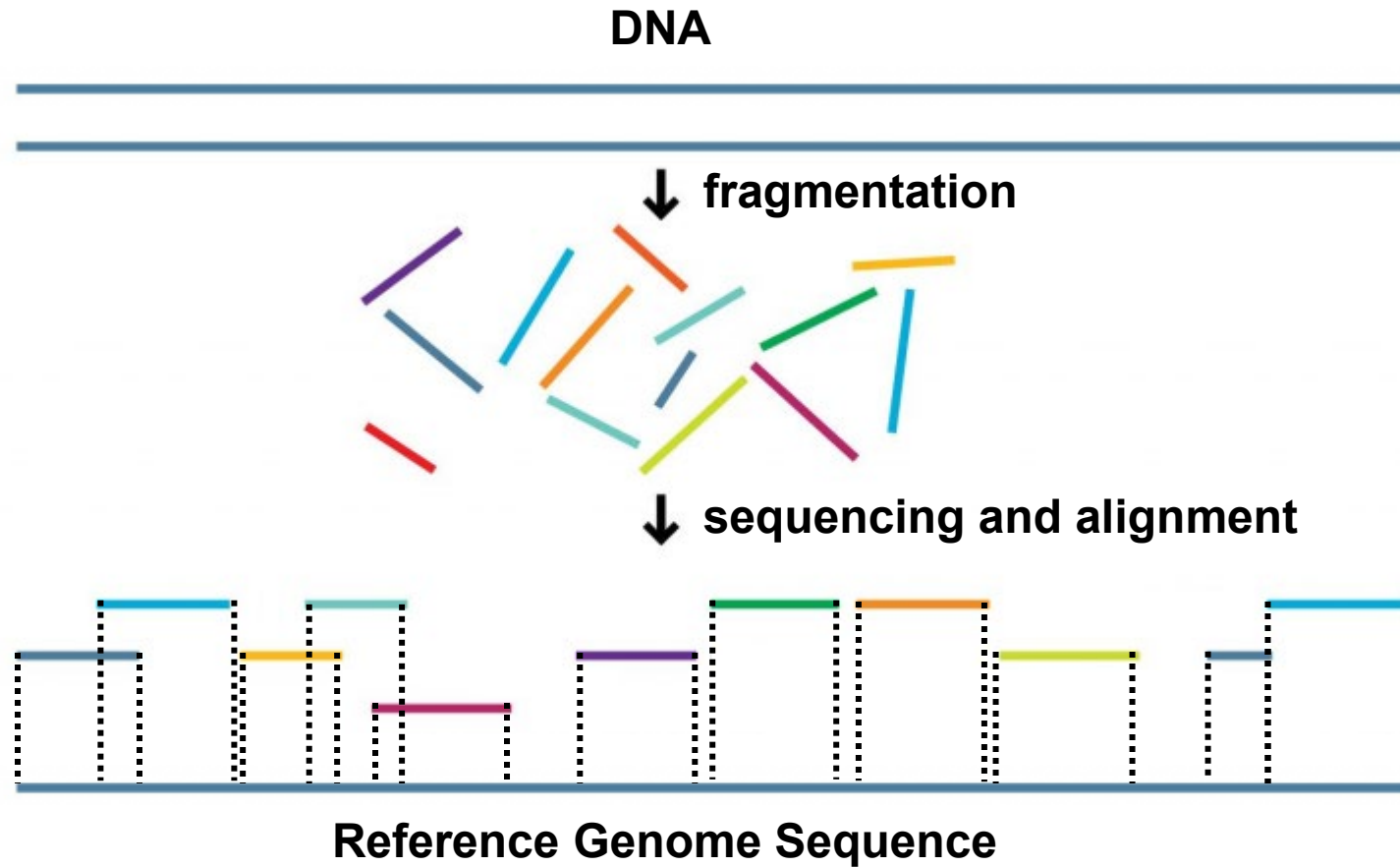
# Course information

- Website hosted on HuskyCT
- Collaboration hours are on Wed. 5:30PM-7:00PM and Wednesday 3:30PM-5:00PM in ITE 230 (starts next week)
- no required textbook to purchase
  - We will use open access books and research articles
- Requirements
  - Reading (Perusall) – 10%
  - 2-3 homeworks/project updates – 40%
  - Final project + presentation – 50%

# Course information

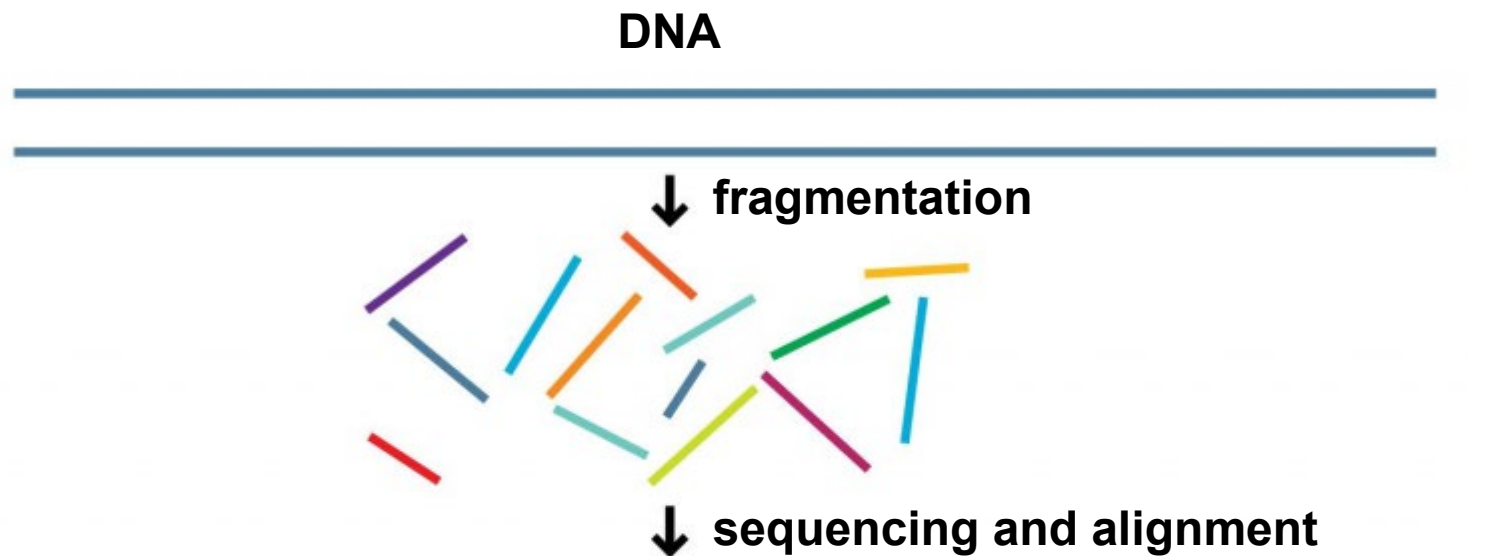
- If you need more time on assignments due to some hardship, let me know.
- research writing guide
- Course survey
- Discord + Ask bot
- Audit/access to huskyct

# Haplotype Assembly



Ploidy (number of  
genome copies)  
 $p = 2$

# Haplotype Assembly



Ploidy (number of  
genome copies)  
 $p = 2$

|   |   |   |   |   |  |  |  |   |   |   |   |  |  |   |   |  |
|---|---|---|---|---|--|--|--|---|---|---|---|--|--|---|---|--|
| 0 | 0 | 0 | 1 | 0 |  |  |  | 1 | 0 | 1 | 0 |  |  | 0 | 0 |  |
| 0 | 1 |   | 0 | 0 |  |  |  | 0 | 1 |   |   |  |  | 0 | 0 |  |
|   |   |   | 1 | 1 |  |  |  |   |   |   |   |  |  |   |   |  |

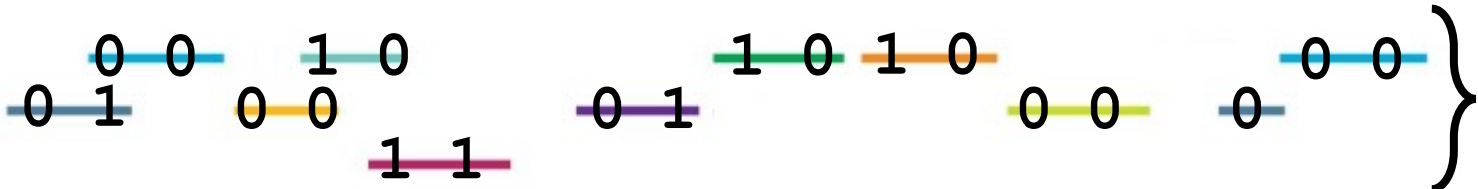
Observed  
(aligned sequence reads)

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 1 | 1 | 1 | 1 | 0 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 0 | 0 | 0 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 1 | 0 | 1 | 1 | 0 | 1 | 0 | 0 |

Unobserved  
(haplotypes)

# Haplotype Assembly

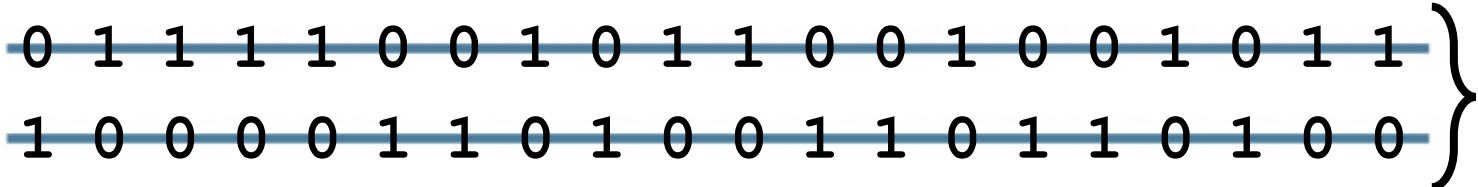
**Given:**



The observed reads are represented by colored horizontal bars, each containing a sequence of 0s and 1s. The colors of the bars are: light blue, yellow, pink, purple, green, orange, light green, and dark blue. The sequences are: 010, 001, 11, 01, 1010, 00, and 000.

Observed  
(aligned  
sequence reads)

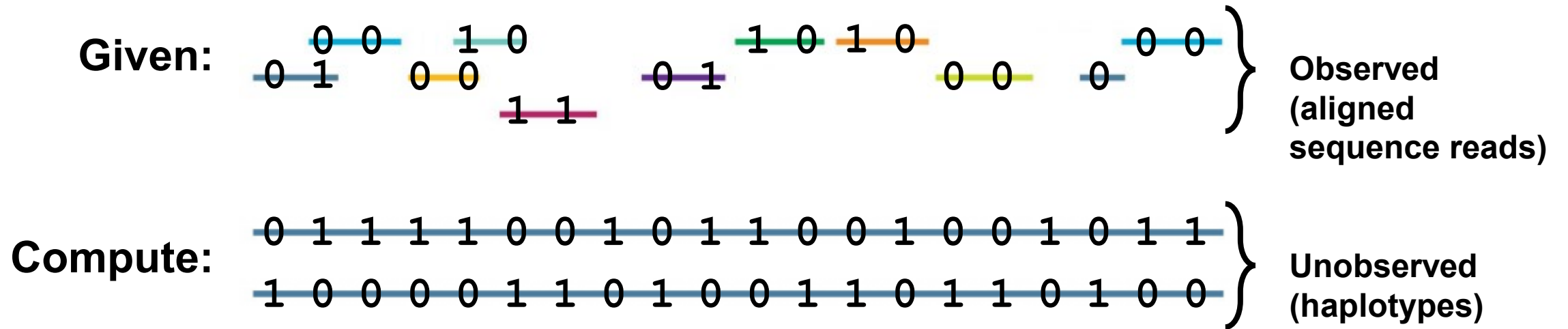
**Compute:**



The unobserved haplotypes are represented by two rows of 0s and 1s. The top row is 01111001011001001011 and the bottom row is 10000110100110110100.

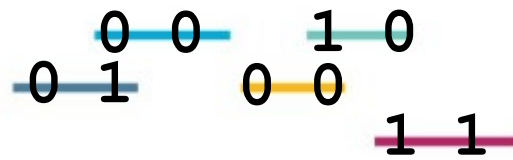
Unobserved  
(haplotypes)

# Haplotype Assembly

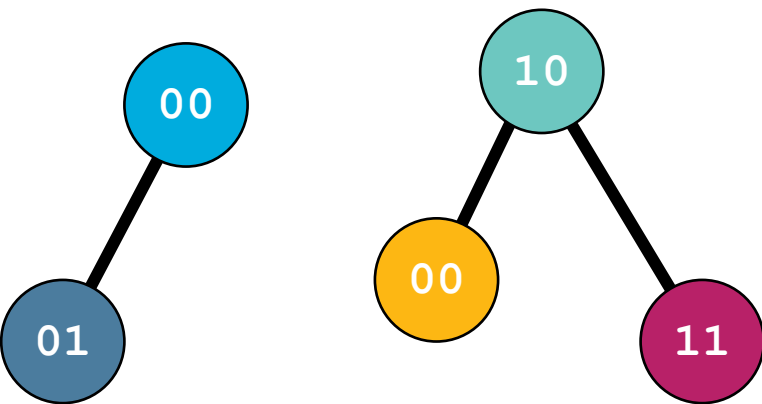


Algorithmic or probabilistic?

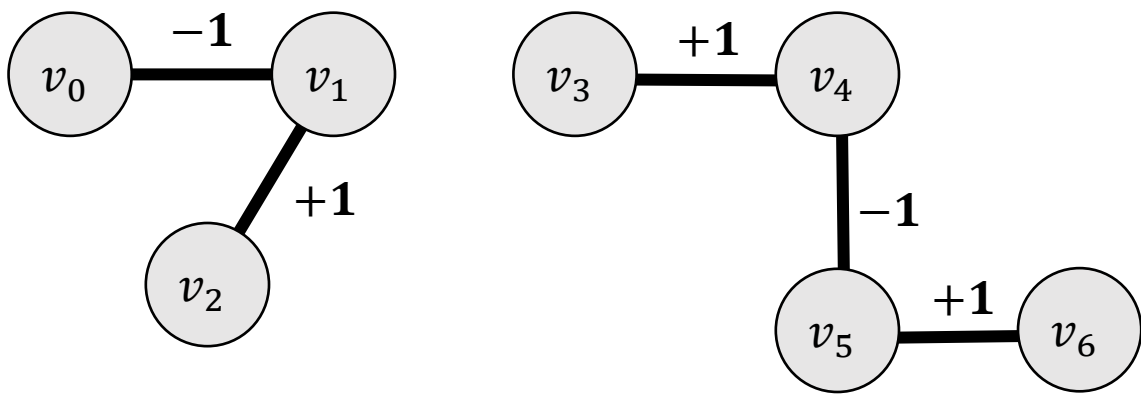
# Algorithmic or probabilistic?



Fragment conflict graph



Compass graph



Between 2 variants,  
∃ only 2 phasings

*cis*

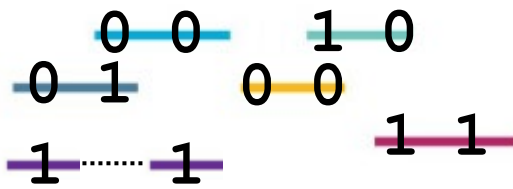
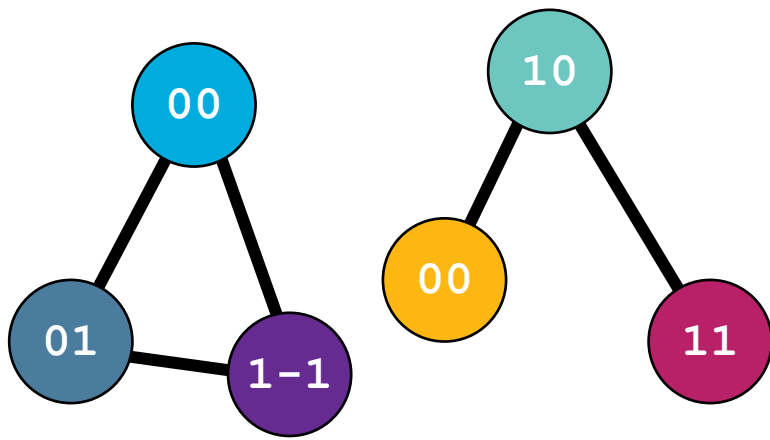
$$\begin{matrix} 0 & 0 \\ 1 & 1 \end{matrix} \rightarrow +1$$

*trans*

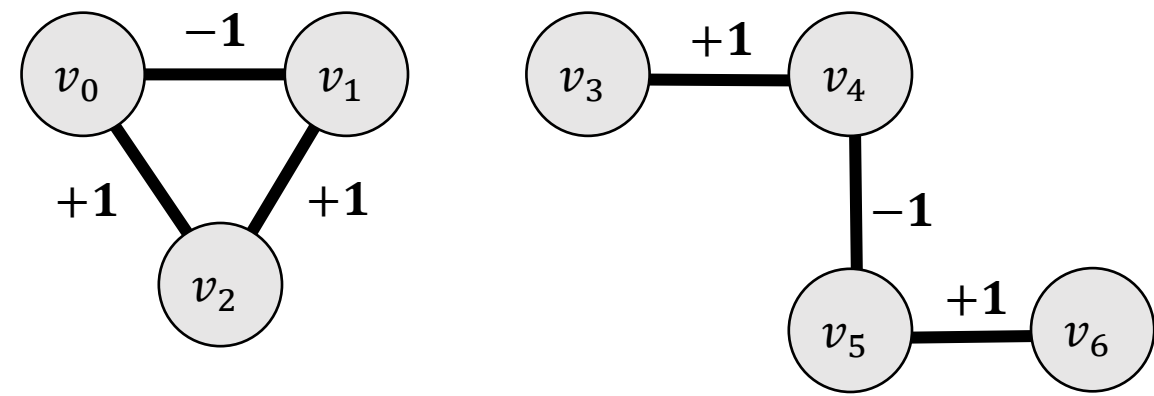
$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \rightarrow -1$$

# Algorithmic or probabilistic?

Fragment conflict graph



Compass graph



Between 2 variants,  
 $\exists$  only 2 phasings

*cis*

$$\begin{matrix} 0 & 0 \\ 1 & 1 \end{matrix} \rightarrow +1$$

*trans*

$$\begin{matrix} 0 & 1 \\ 1 & 0 \end{matrix} \rightarrow -1$$

# Algorithmic or probabilistic?

DNA

Diploid ( $p = 2$ )  
no haplotype intersection  
between different phasings

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 1 |
| 1 | 1 | 1 | 0 |

Polyploid ( $p > 2$ )  
phasings share haplotypes

|   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | 1 | 1 | 1 |
| 0 | 0 | 1 | 0 | 0 | 0 | 0 | 1 | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 | 0 | 1 | 1 | 0 | 0 | 0 | 0 | 0 | 1 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |

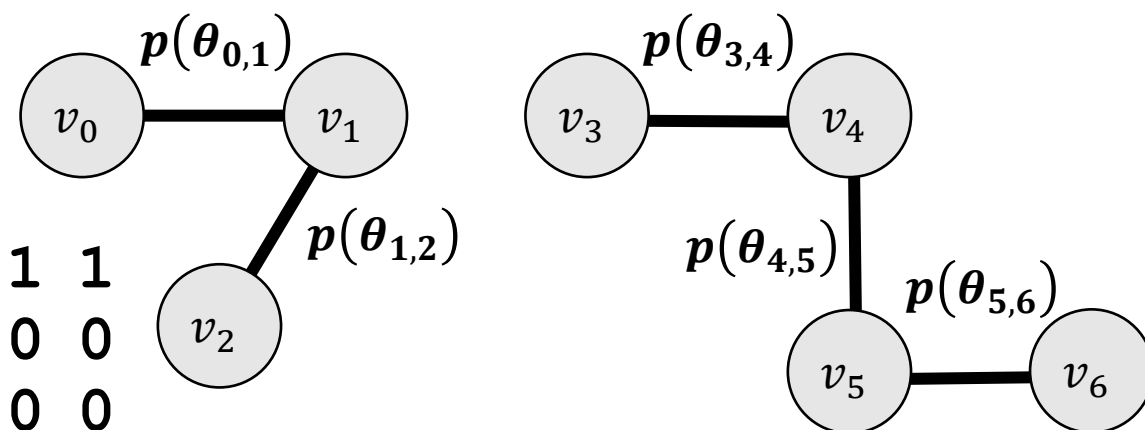


ploidy  $p = 4$



ploidy  
 $p = 2, 4, 5, 6, 7, 8, \text{ or } 10$

Compass graph



**Question: If I sample the DNA of everyone in this room and looked at the sequences of DNA, what would I find?**

**Question: If I sample the DNA of everyone in this room and looked at the sequences of DNA, what would I find?**

**genome variants**

**individuals**

CGACCCCTCCCAGTGGAGCCAG  
CGACCCCTCCCAGTGGAGCCAG  
CGACACTATCAATAGAGCCAG  
CGACACTATCAATAGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGAACGAGGCAGA  
AACTCTGATGAACGGAAACAG

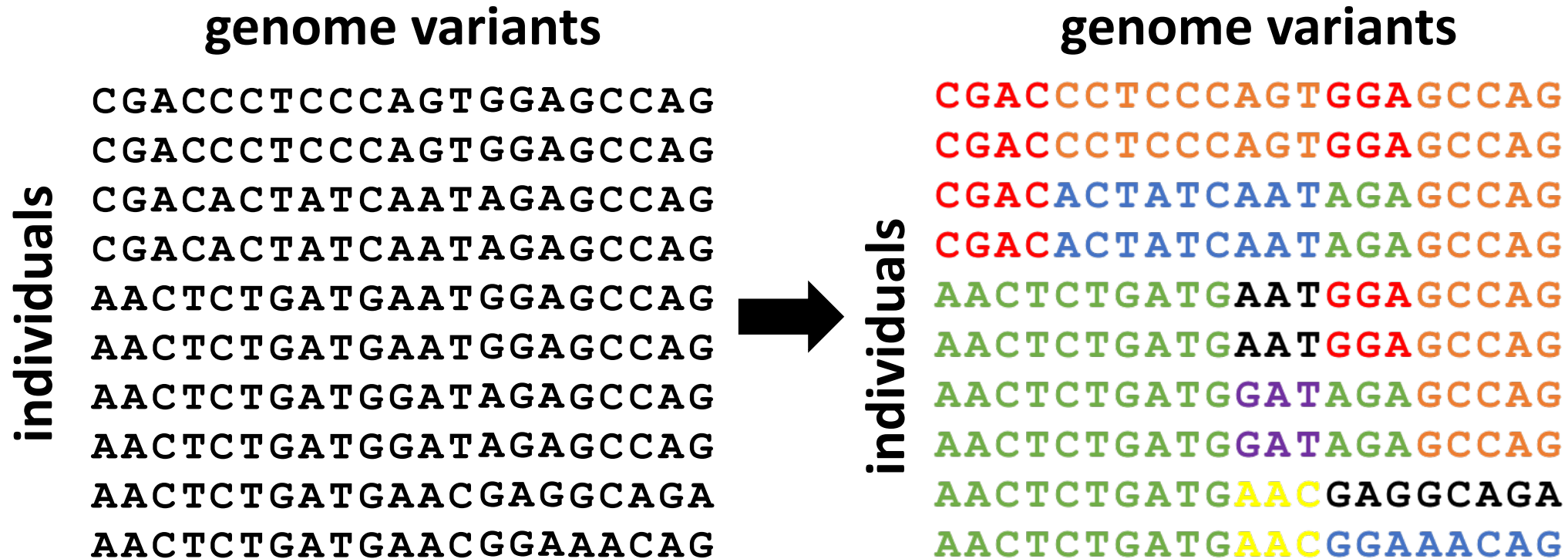
**Problem: 'color' pieces of sequence based on shared ancestry**

## **genome variants**

**individuals**

```
CGACCCCTCCCAGTGGAGCCAG
CGACCCCTCCCAGTGGAGCCAG
CGACACTATCAATAGAGCCAG
CGACACTATCAATAGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGAATGGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGGATAGAGCCAG
AACTCTGATGAACGAGGCAGA
AACTCTGATGAACGGAAACAG
```

Problem: 'color' pieces of sequence based on shared ancestry



How could you do this?

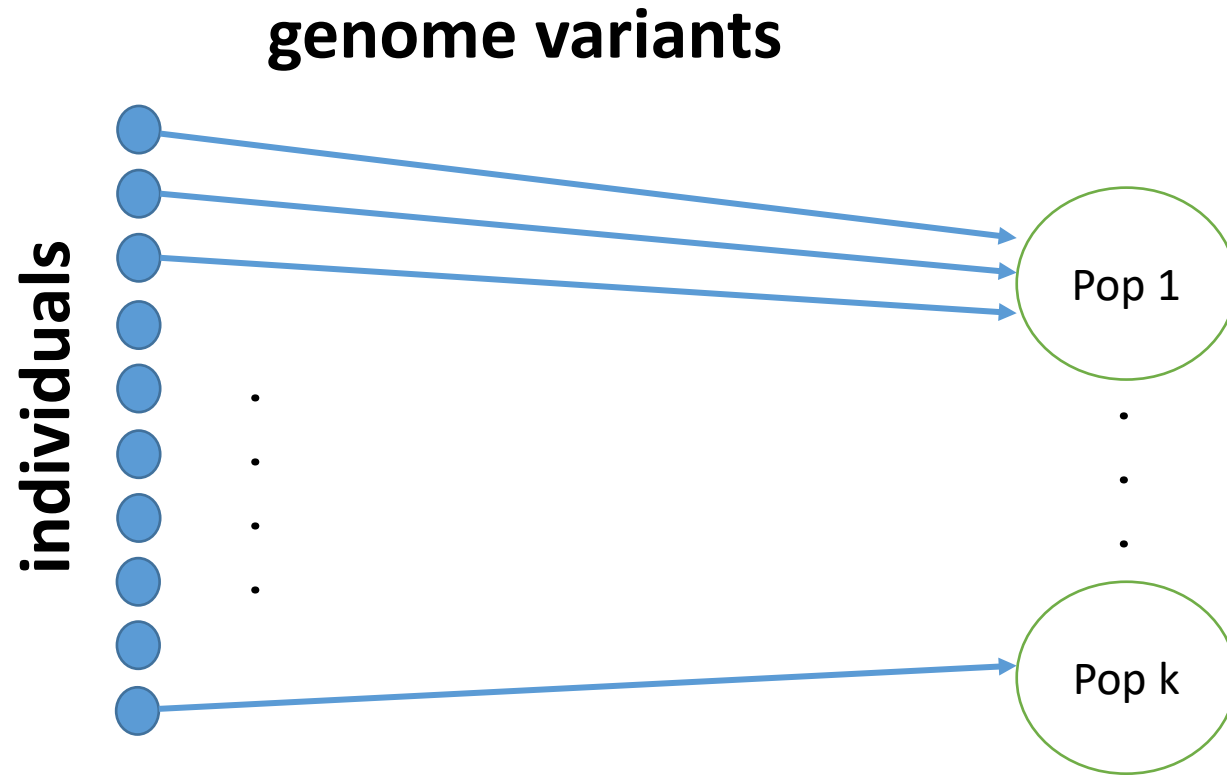
**Idea: what if we assumed that each variant was sampled from a population?**

## genome variants

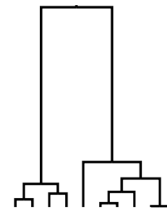
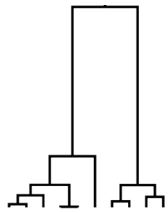
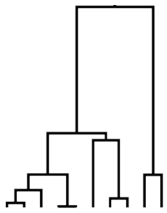
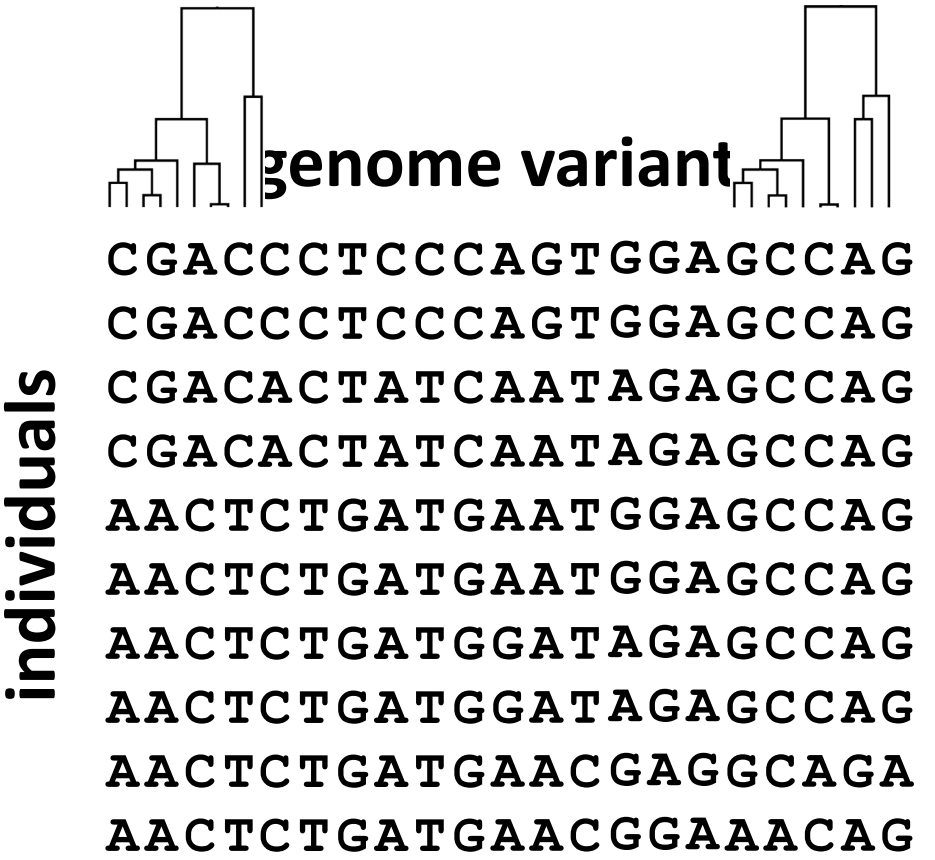
# individuals

C  
C  
C  
C  
A  
A  
A  
A  
A  
A

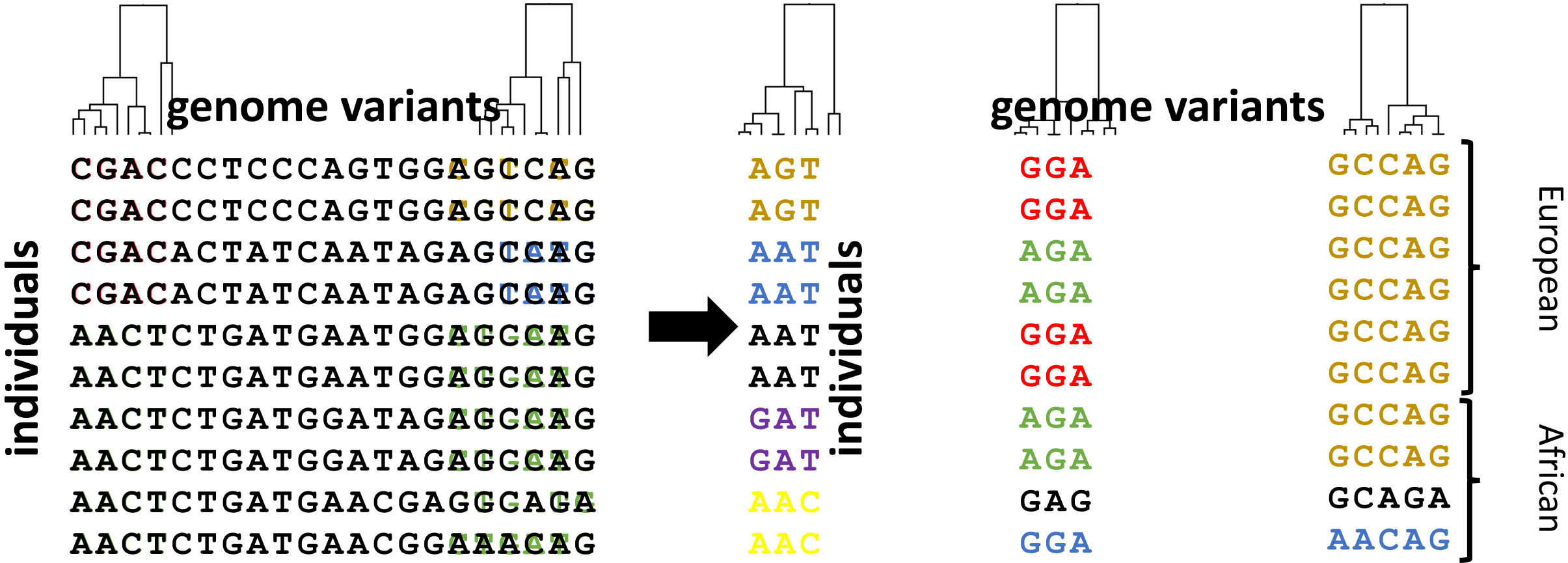
Idea: what if we assumed that each variant was sampled from a population?



# Probabilistic models

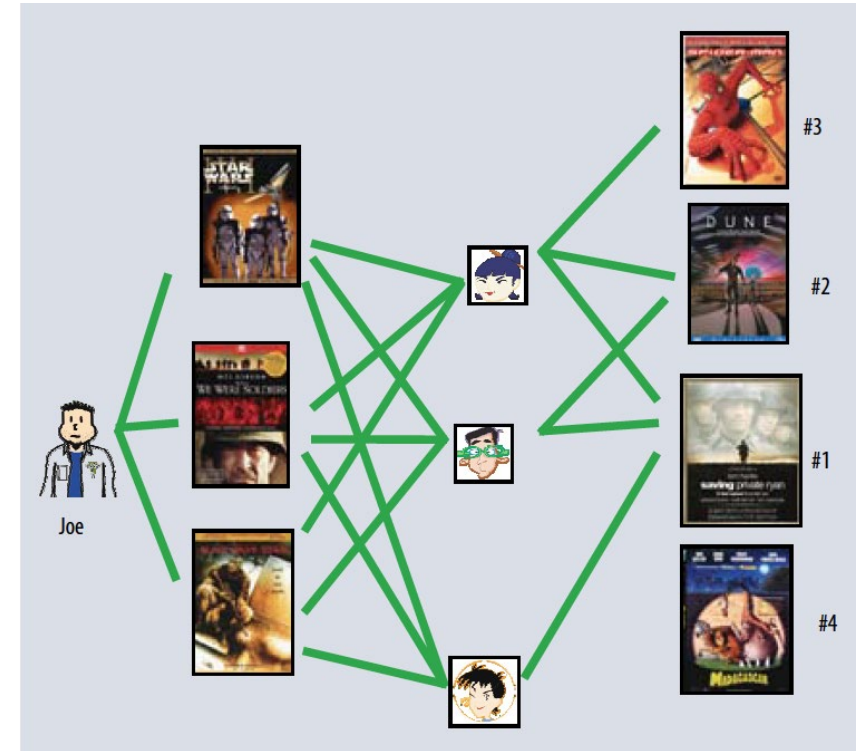


# Probabilistic models

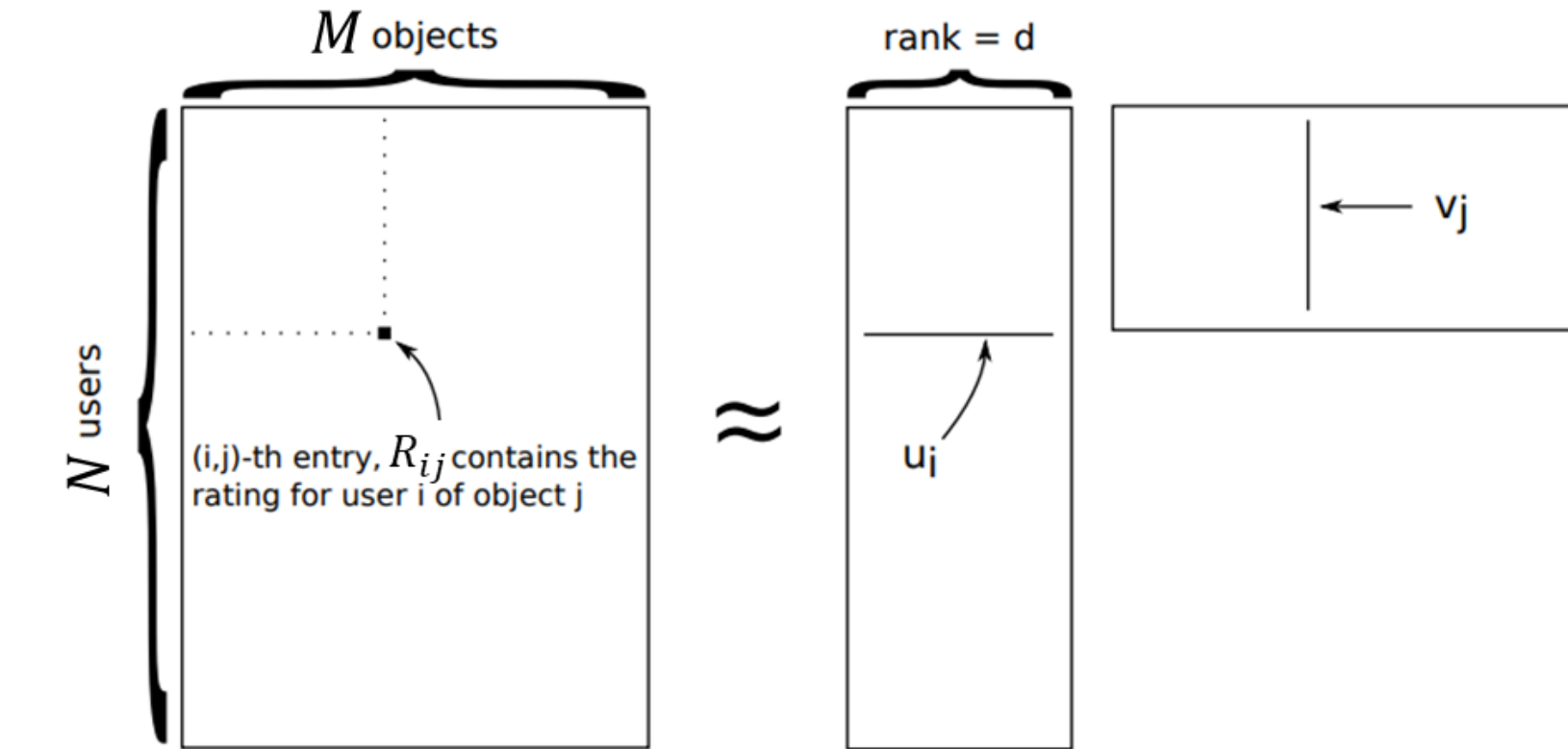


# Recommendation systems

- Question: how do we get more people watching Netflix?
- Problem: recommend movies to users that they are likely to finish watching and rate highly
- Given a set of users with their previous ratings for a set of movies, can we predict the rating they will assign to a movie they have not previously rated?
- Netflix was interesting in this problem, offered a one million dollar prize

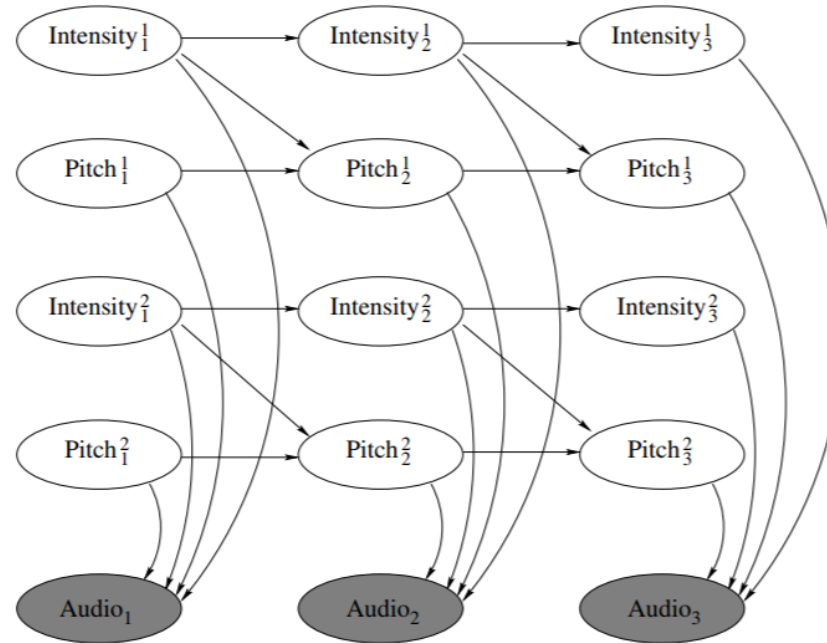


# Recommendation systems



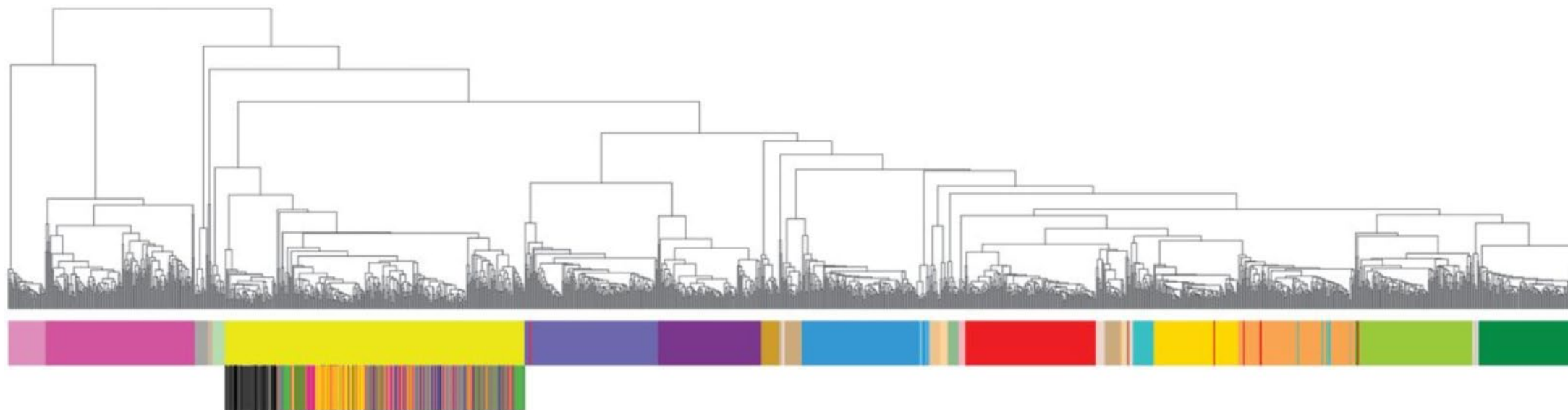
# Lower dimensional embedding

# Musical transcription



**Fig. 1.** A DGM for two-instrument transcription. Shaded nodes are observed data; unshaded nodes are “hidden” or “latent” variables.

## A Gene expression clustering



### All tissues (main tissues in bold)

|                       |                |               |                     |
|-----------------------|----------------|---------------|---------------------|
| <b>Adipose tissue</b> | Colon          | <b>Lung</b>   | Prostate            |
| Adrenal gland         | Esophagus      | <b>Muscle</b> | Reproductive organs |
| <b>Artery</b>         | Fibroblasts    | <b>Nerve</b>  | <b>Skin</b>         |
| <b>Blood</b>          | <b>Heart</b>   | Ovary         | Stomach             |
| Brain                 | Kidney / Liver | Pancreas      | Testis              |
| Breast                | LCL            | Pituitary     | <b>Thyroid</b>      |

# Disparate models, common theme

- Each model developed in its own community
- What is common to all of these models?
- Recent work in ML has focused on developing tools to create, analyze, and implement these models and inference algorithms generally
- This has led to the field of **Bayesian machine learning**

# Course description

Probabilistic modelling

Posterior inference

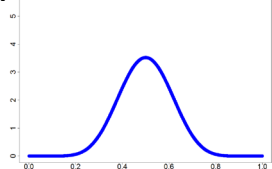
Model criticism

# Bayesian ML workflow

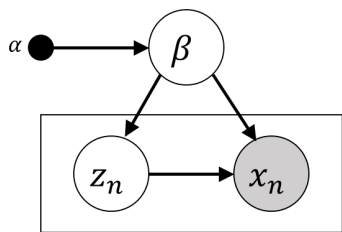
domain expertise



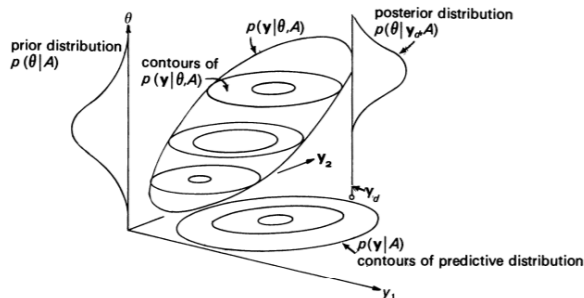
prior knowledge



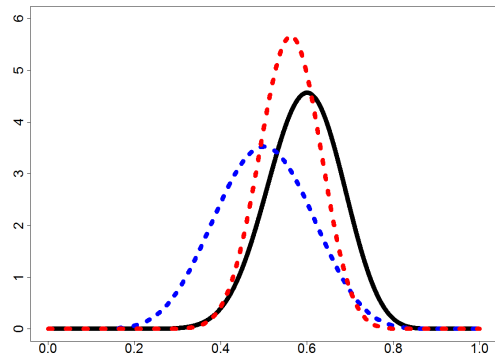
Probabilistic modeling



model criticism



Posterior inference



predictions & exploration

CGACCCTCCCAGTGGAGCCAG  
CGACCCTCCCAGTGGAGCCAG  
CGACACTATCAATAGAGCCAG  
CGACACTATCAATAGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGAACGAGGCAGA  
AACTCTGATGAACGGAAACAG

data

CGACCCTCCCAGTGGAGCCAG  
CGACCCTCCCAGTGGAGCCAG  
CGACACTATCAATAGAGCCAG  
CGACACTATCAATAGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGAACGAGGCAGA  
AACTCTGATGAACGGAAACAG

# Course description

## Probabilistic modelling

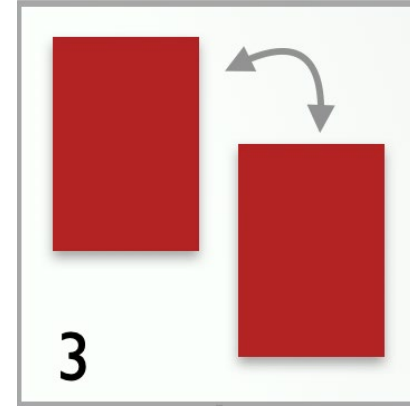
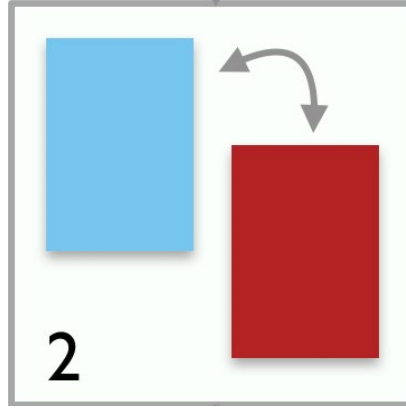
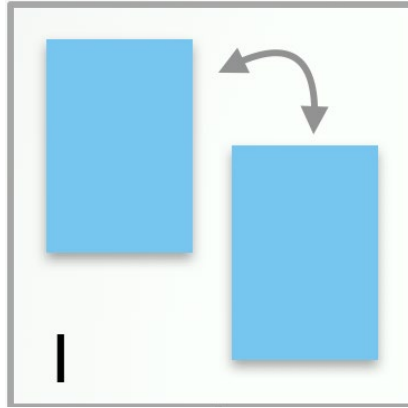
**Goal 1:** Develop a formal mathematical framework for probabilistic modelling.

Posterior inference

Model criticism

# A simple 3 card problem

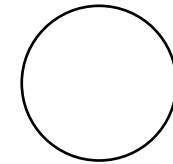
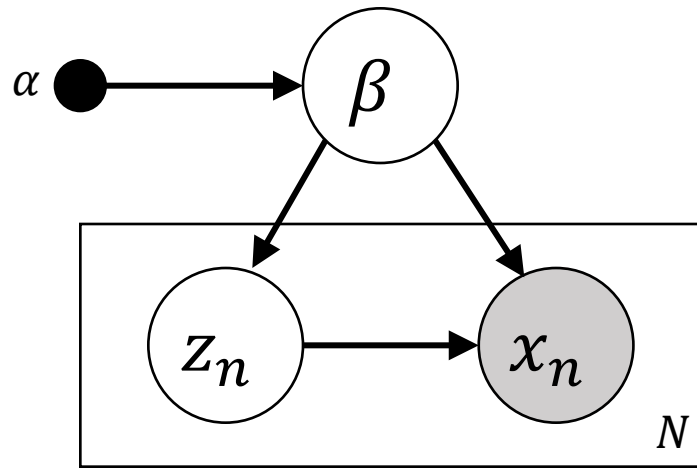
- 3 cards with colored sides, blue-blue, blue-red, red-red



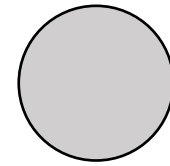
- Pick a card uniformly at random and show a side at random.
- E.g., say we do this and we see blue. What is the probability the other side is blue?

# Probabilistic modelling

- Graphical models capture assumptions about the data generation process



Random variable

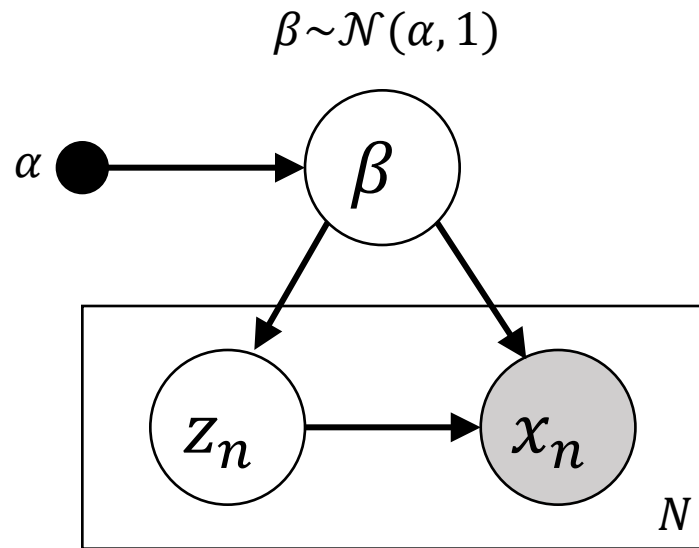


Observed (random) data

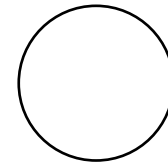


Hyperparameters

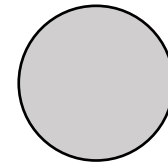
# Probabilistic modelling



- Graphical models capture assumptions about the data generation process



Random variable



Observed (random) data



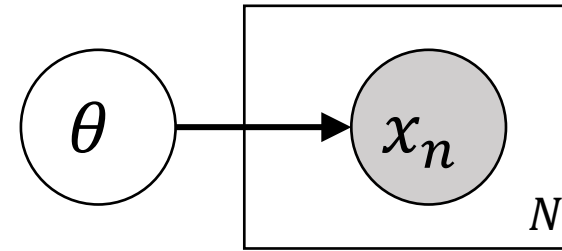
Hyperparameters

# Modelling the flipping of a coin

- $n$  coin flips
- Heads or tails is recorded
- What are the:
  - Latent random variables
  - Observed random variables
  - Dependencies

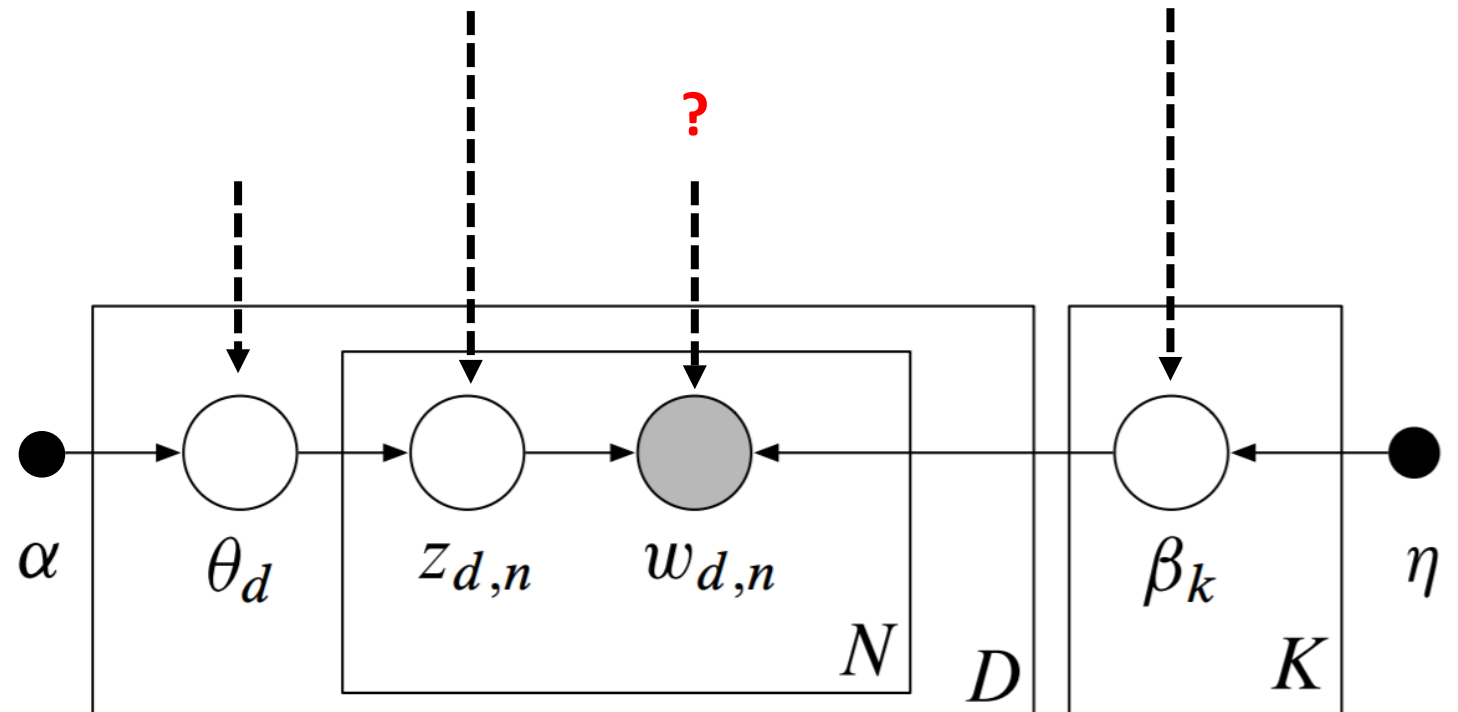
# Modelling the flipping of a coin

- $n$  coin flips
- Heads or tails is recorded
- What are the:
  - Latent random variables
  - Observed random variables
  - Dependencies



# STRUCTURE (an approximation)

- How do we model shared genetic ancestry?
- $N$ =number of variants
- $D$ =number of individuals
- $K$ =number of populations



# STRUCTURE (an approximation)

- How do we model shared genetic ancestry?
- $N$ =number of variants
- $D$ =number of individuals
- $K$ =number of populations

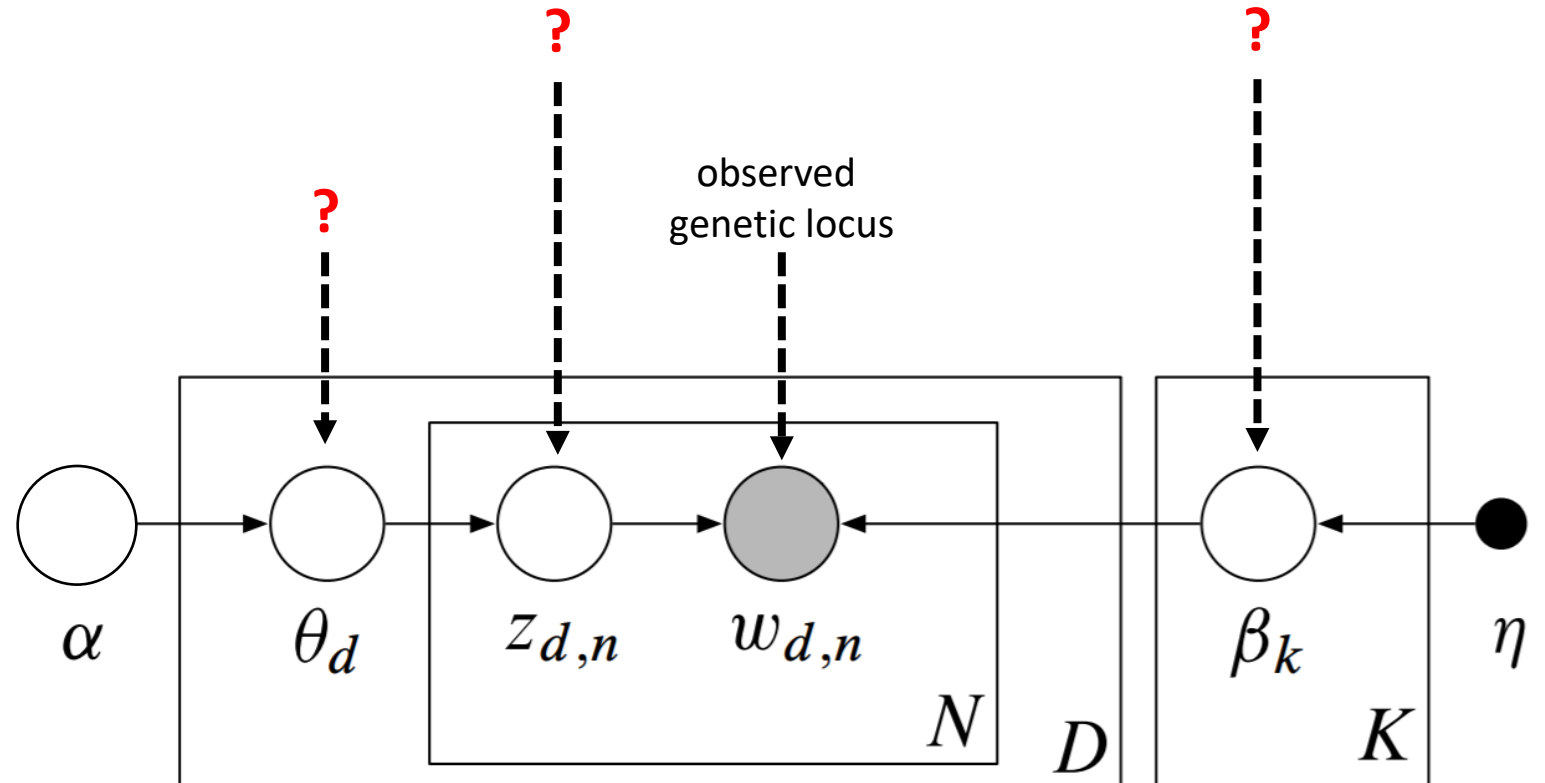
**genome variants**

**individuals**

|  |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |   |
|--|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
|  | C | G | A | C | C | T | C | C | C | A | G | T | G | G | A | G | C | C | A | G |   |
|  | C | G | A | C | C | T | C | C | C | A | G | T | G | G | A | G | C | C | A | G |   |
|  | C | G | A | C | A | C | T | A | T | C | A | A | T | A | G | A | G | C | C | A | G |
|  | C | G | A | C | A | C | T | A | T | C | A | A | T | A | G | A | G | C | C | A | G |
|  | A | A | C | T | C | T | G | A | T | G | A | A | T | G | G | A | G | C | C | A | G |
|  | A | A | C | T | C | T | G | A | T | G | A | A | T | G | G | A | G | C | C | A | G |
|  | A | A | C | T | C | T | G | A | T | G | G | A | T | A | G | A | G | C | C | A | G |
|  | A | A | C | T | C | T | G | A | T | G | G | A | T | A | G | A | G | C | C | A | G |
|  | A | A | C | T | C | T | G | A | T | G | A | A | C | G | A | G | G | C | A | G | A |
|  | A | A | C | T | C | T | G | A | T | G | A | A | C | G | G | A | A | A | C | A | G |

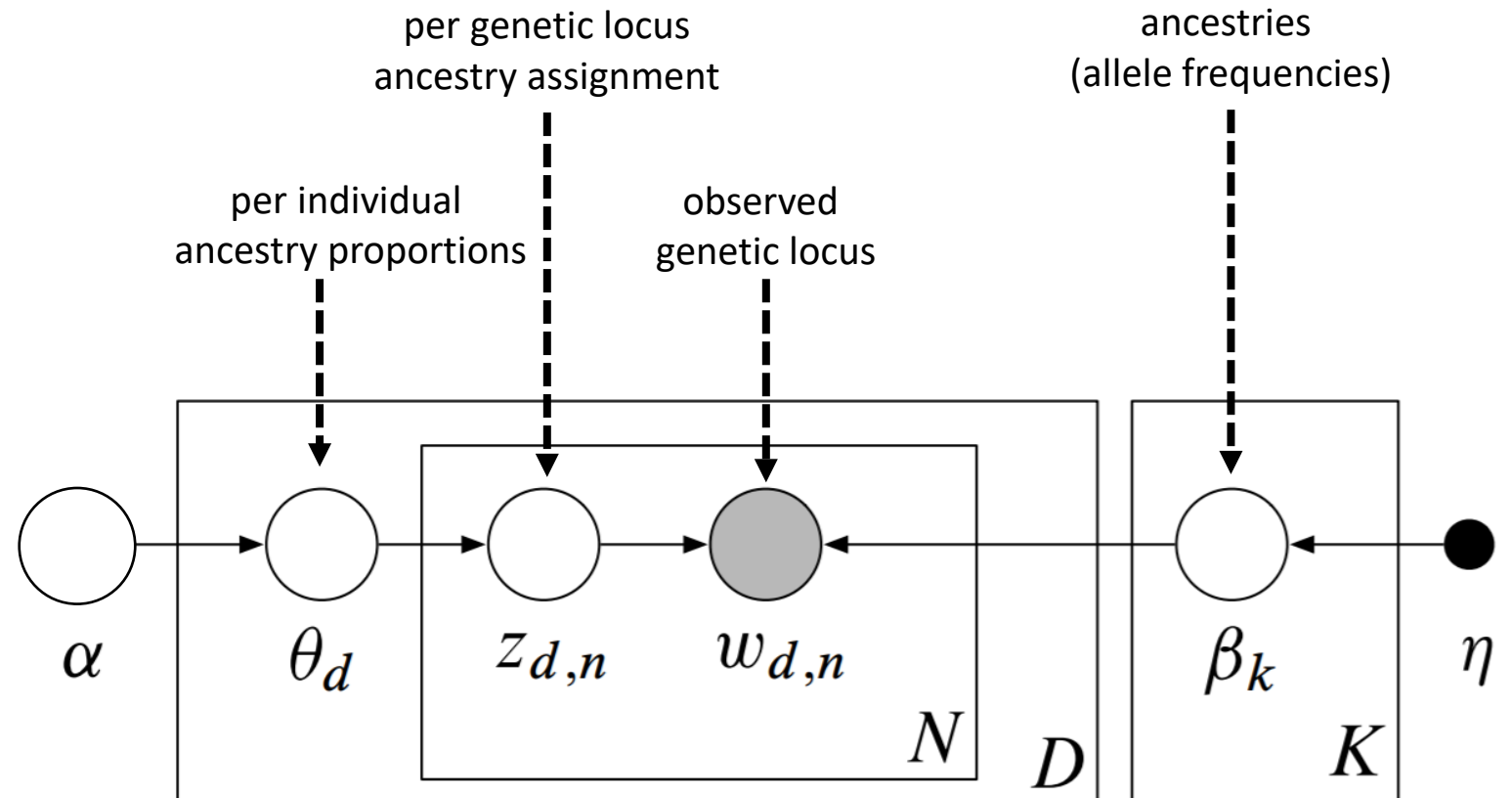
# STRUCTURE (an approximation)

- How do we model shared genetic ancestry?
- $N$ =number of variants
- $D$ =number of individuals
- $K$ =number of populations



# STRUCTURE (an approximation)

- How do we model shared genetic ancestry?
- $N$ =number of variants
- $D$ =number of individuals
- $K$ =number of populations



# Course description

Probabilistic modelling

**Posterior inference**

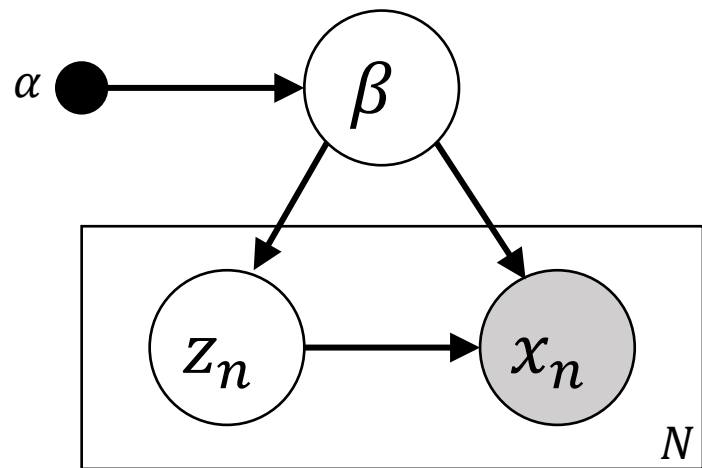
**Goal 2:** Build efficient inference algorithms for large models and data.

Model criticism

# Inference

**Inference** requires reasoning about the value of unobserved (latent) variables (e.g.,  $\beta$ ,  $\mathbf{z}$ ) given observations (e.g.,  $\mathbf{x}$ ) which are related to the latent variables but insufficient to fully determine them.

This requires the notion of **uncertainty**.



$$p(x, z, \beta) = p(z, \beta | x) p(x)$$

$$p(z, \beta | x) = \frac{p(x, z, \beta)}{p(x)}$$

# Bayesian inference

## Bayes Theorem

$$\begin{array}{c} \text{posterior} \\ p(z, \beta | x) \end{array} = \frac{\begin{array}{c} \text{data likelihood} \\ p(x | z, \beta) \end{array} \begin{array}{c} p(z, \beta) \\ \text{prior} \end{array}}{\begin{array}{c} \int \int p(x, z, \beta) dz d\beta \\ \text{marginal likelihood} \\ \text{or model evidence} \end{array}}$$

- Model evidence often intractable to evaluate

# Intractability of computing model evidence

- Exact methods
- Approximate methods
- Sampling methods

posterior

$$p(\theta|x)$$

=

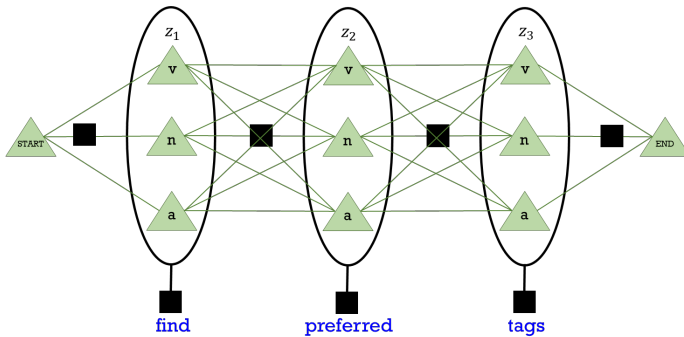
$$\frac{1}{\int p(x, \theta) d\theta} * p(x, \theta)$$

joint density  
(data likelihood \* prior)

normalizing constant  
model evidence

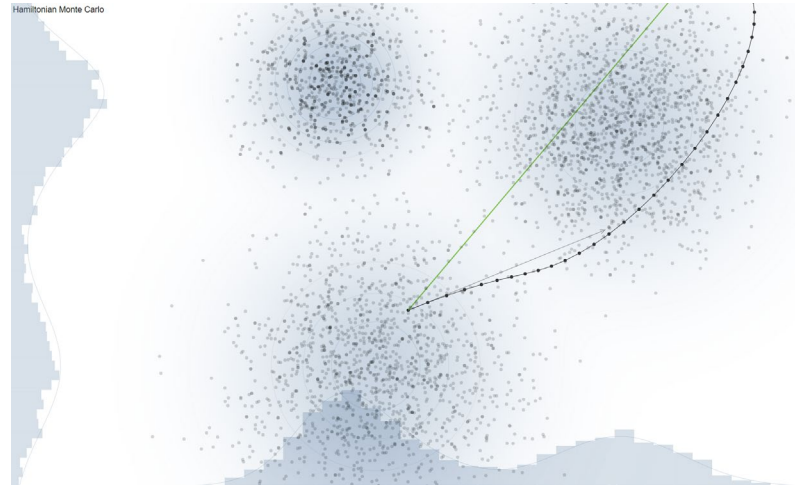
# Approaches to Inference

## Exact



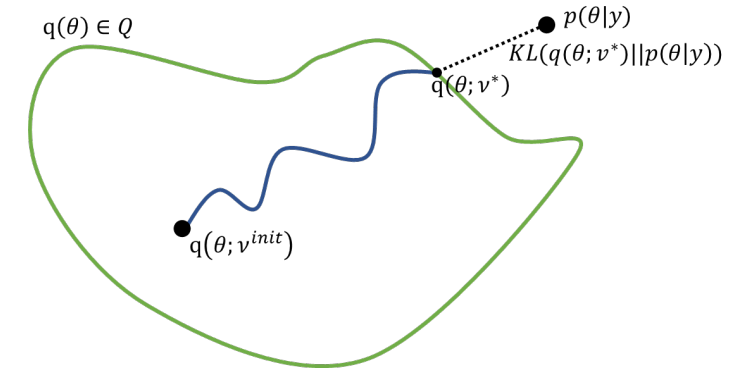
Only available for  
few models

## Sampling



Strong theoretical  
guarantees, typically easy  
to derive, difficult to run

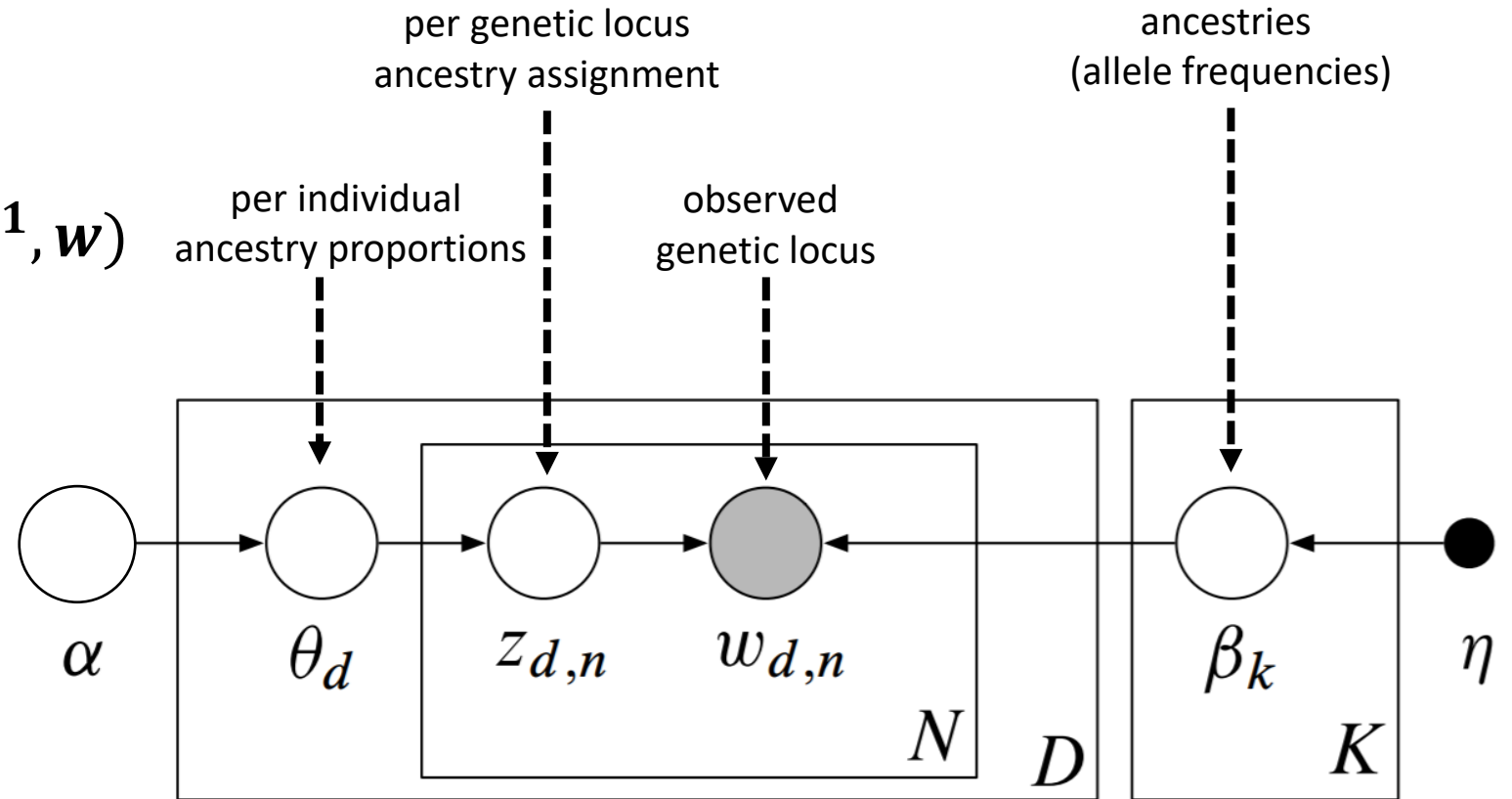
## Optimization



Faster than sampling,  
but approximate,  
many assumptions

# STRUCTURE (inference)

- Originally, MCMC
  - Sample  $\theta^i, z^i$  from  $P(\theta, z | \beta^{i-1}, w)$
  - Sample  $\beta$  from  $P(\beta | z^i, \theta^i, w)$
  - Update  $\alpha$  using a Metropolis-Hastings step
- 14 years later, VI approach
  - Raj *et al.* Genetics, 2014.



# Course description

Probabilistic modelling

Posterior inference

**Model criticism**

**Goal 3:** Develop methods for criticizing our models and interpreting results.

# Model criticism

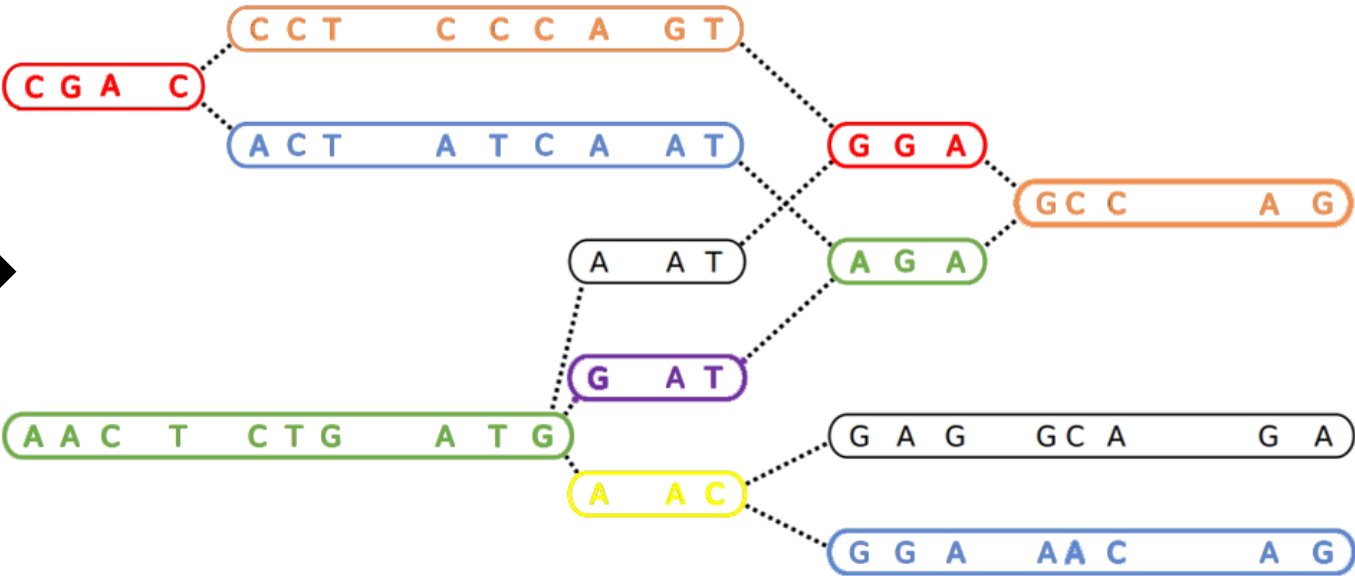
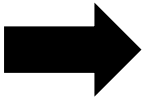
- Examples of prediction
  - Future stock prices based on market history
  - Topic distribution for next year's NYT articles
  - Population distribution for new samples of genetic material
- Model criticism attempts to find where the model falls short so we can iterate.
  - Cross validation
  - Posterior predictive checks

# Model criticism examples

## genome variants

individuals

CGACCCTCCCAGTGGAGCCAG  
CGACCCTCCCAGTGGAGCCAG  
CGACACTATCAATAGAGCCAG  
CGACACTATCAATAGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGAACGAGGCAGA  
AACTCTGATGAACGGAAACAG



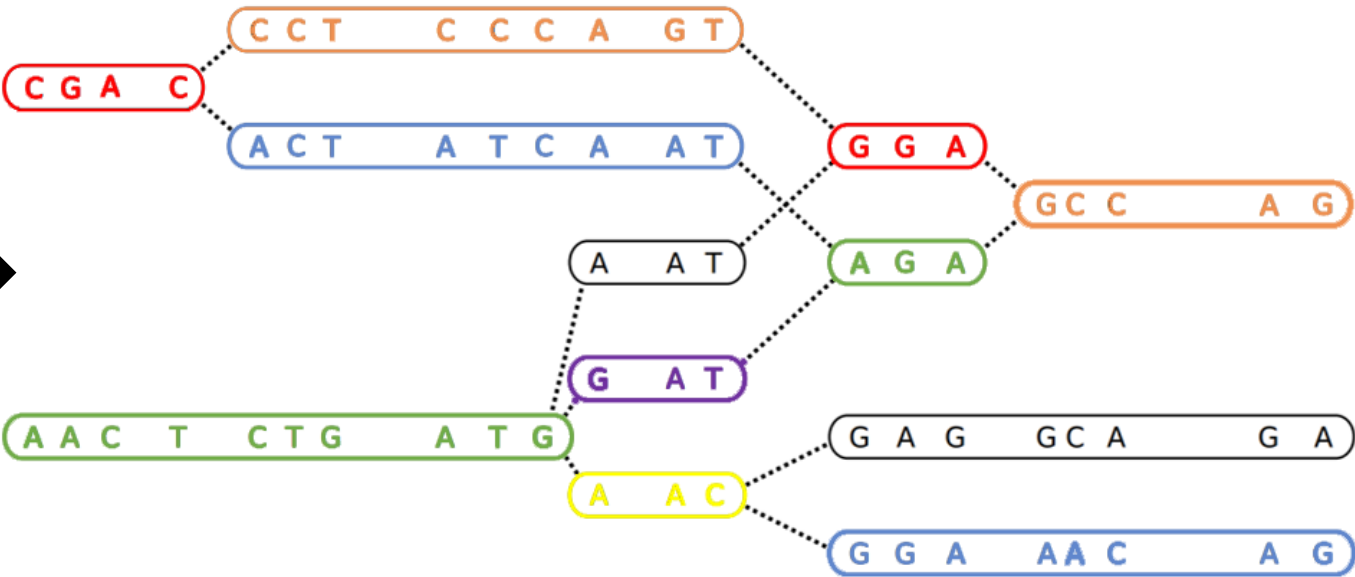
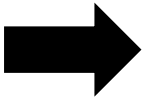
# Prediction example

Where does the new unlabeled **AACTCTGATGAACGGAAACAG** cluster?

## genome variants

individuals

CGACCCTCCCAGTGGAGCCAG  
CGACCCTCCCAGTGGAGCCAG  
CGACACTATCAATAGAGCCAG  
CGACACTATCAATAGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGAATGGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGGATAGAGCCAG  
AACTCTGATGAACGAGGCAGA  
AACTCTGATGAACGGAAACAG



# individuals

# individuals

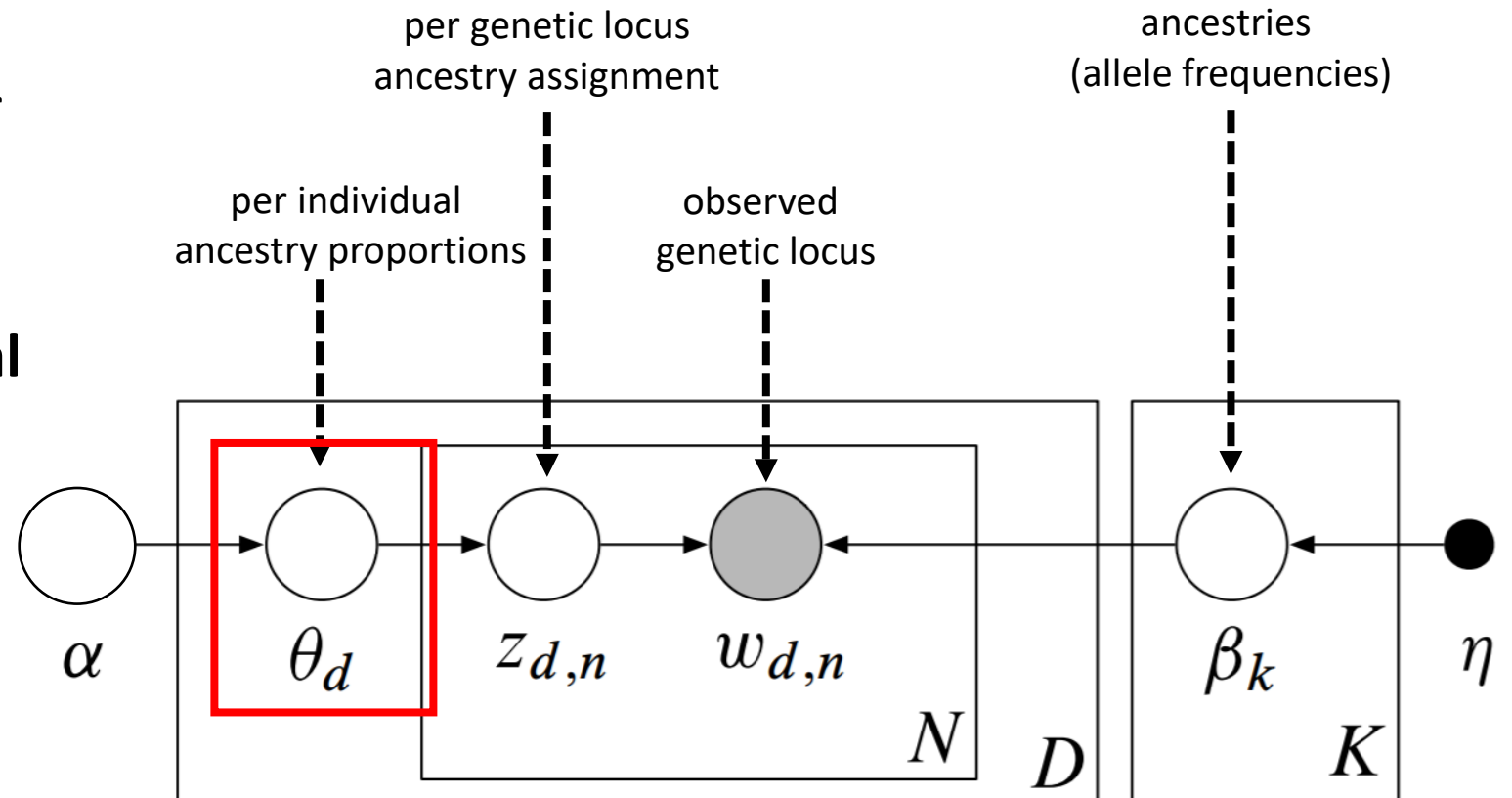
# individuals

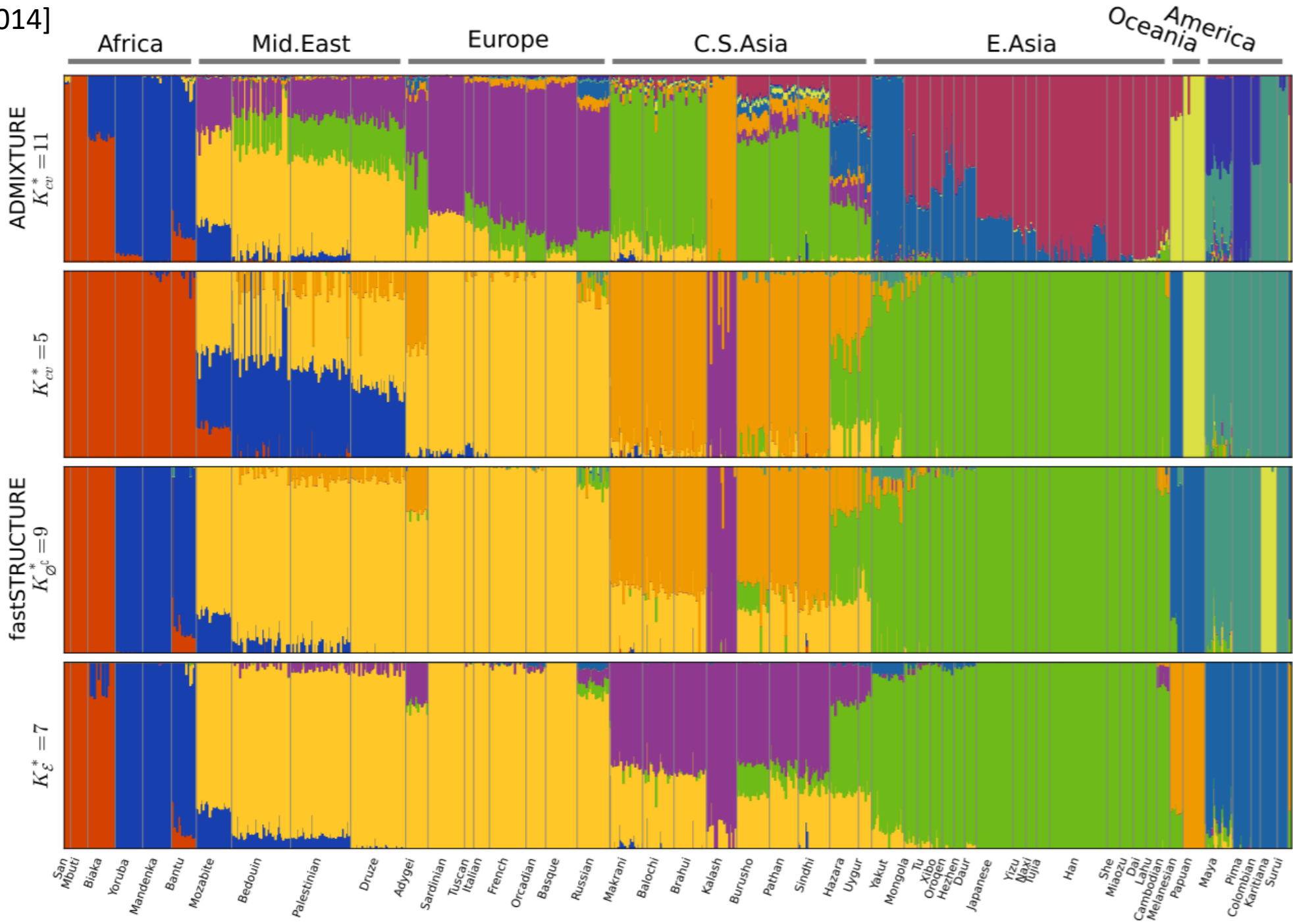
# individuals



# STRUCTURE (interpretation)

- What are some of the r.v.s useful for interpretation of genetic history?
  - $\theta$  gives the mixture of ancestry for each individual
  - $z$  gives local ancestry information for a genetic locus
  - $\beta$  gives the ancestry distribution





# A similar problem, different area

- Topic modeling
- Problem: Given a corpus of articles, figure out the latent topics and which articles express which topics



# Archaeogenomic evidence from the southwestern US points to a pre-Hispanic scarlet macaw breeding colony

Richard J. George<sup>a</sup>, Stephen Plog<sup>b,1</sup>, Adam S. Watson<sup>c</sup>, Kari L. Schmidt<sup>d,e</sup>, Brendan J. Culleton<sup>a,f</sup>, Thomas K. Harper<sup>a</sup>, Patricia A. Gilman<sup>g</sup>, Steven A. LeBlanc<sup>h</sup>, George Amato<sup>e</sup>, Peter Whiteley<sup>c</sup>, Logan Kistler<sup>i</sup>, and Douglas J. Kennett<sup>a,f,1</sup>

<sup>a</sup>Department of Anthropology, The Pennsylvania State University, University Park, PA 16802; <sup>b</sup>Department of Anthropology, University of Virginia, Charlottesville, VA 22904; <sup>c</sup>Division of Anthropology, American Museum of Natural History, New York, NY 10024; <sup>d</sup>Department of Ecology, Evolution and Environmental Biology, Columbia University, New York, NY 10027; <sup>e</sup>Sackler Institute for Comparative Genomics, American Museum of Natural History, New York, NY 10024; <sup>f</sup>Institutes for Energy and the Environment, The Pennsylvania State University, University Park, PA 16802; <sup>g</sup>Department of Anthropology, University of Oklahoma, Norman, OK 73019; <sup>h</sup>Peabody Museum of American Archaeology and Ethnology, Harvard University, Cambridge, MA 02138; and <sup>i</sup>Department of Anthropology, Smithsonian Institution, Washington, DC 20560

Edited by James A. Brown, Northwestern University, Evanston, IL, and approved July 13, 2018 (received for review April 4, 2018)

Hundreds of scarlet macaw (*Ara macao cyanoptera*) skeletons have been recovered from archaeological contexts in the southwestern United States and northwestern Mexico (SW/NW). The location of these skeletons, >1,000 km outside their Neotropical endemic range, has suggested a far-reaching pre-Hispanic acquisition network. Clear evidence for scarlet macaw breeding within this network is only known from the settlement of Paquimé in NW dating between 1250 and 1450 CE. Although some scholars have speculated on the probable existence of earlier breeding centers in the SW/NW region, there has been no supporting evidence. In this study, we performed an ancient DNA analysis of scarlet macaws recovered from archaeological sites in Chaco Canyon and the contemporaneous Mimbres area of New Mexico. All samples were directly radiocarbon dated between 900 and 1200 CE. We reconstructed complete or near-complete mitochondrial genome sequences of 14 scarlet macaws from five different sites. We observed remarkably low genetic diversity in this sample, consistent with breeding of a small founder population translocated outside their natural range. Phylogeographic comparisons of our ancient DNA mitogenomes with mitochondrial sequences from macaws collected during the last 200 years from their endemic Neotropical range identified genetic affinity between the ancient macaws and a single rare haplogroup (Haplo6) observed only among wild macaws in Mexico and northern Guatemala. Our results suggest that people at an undiscovered pre-Hispanic settlement dating between 900 and 1200 CE managed a macaw breeding colony outside their endemic range and distributed these symbolically important birds through the SW.

along with other exogenous items and products including cacao, copper bells, marine shell/bracelets, and distinct ceramic vessels, may have both contributed to and were products of emergent sociopolitical complexity in Chaco Canyon (19, 21).

Some scholars have hypothesized that scarlet macaws were directly acquired from lowland tropical regions between 900 and 1200 CE via long-distance treks to the northern extent of their range in the Mexican Gulf Coast state of Tamaulipas, and possibly to more distant ranges in the Isthmus of Tehuantepec and Central America (24). Others have suggested that these birds passed from one community to the next from Mesoamerica to the SW/NW (e.g., refs. 25–27). Alternatively, here we test the hypothesis proposed by Crown (28), that lengthy trips were mitigated by the presence of intermediate breeding centers in the area between the northern endemic range of scarlet macaws and the SW/NW.

Such centers may have raised generations of captive macaws and maintained a stock of age-specific macaws for transport throughout the SW/NW, as occurred at the later (1250–1450 CE) settlement of Paquimé in the Mexican state of Chihuahua. Paquimé was an

## Significance

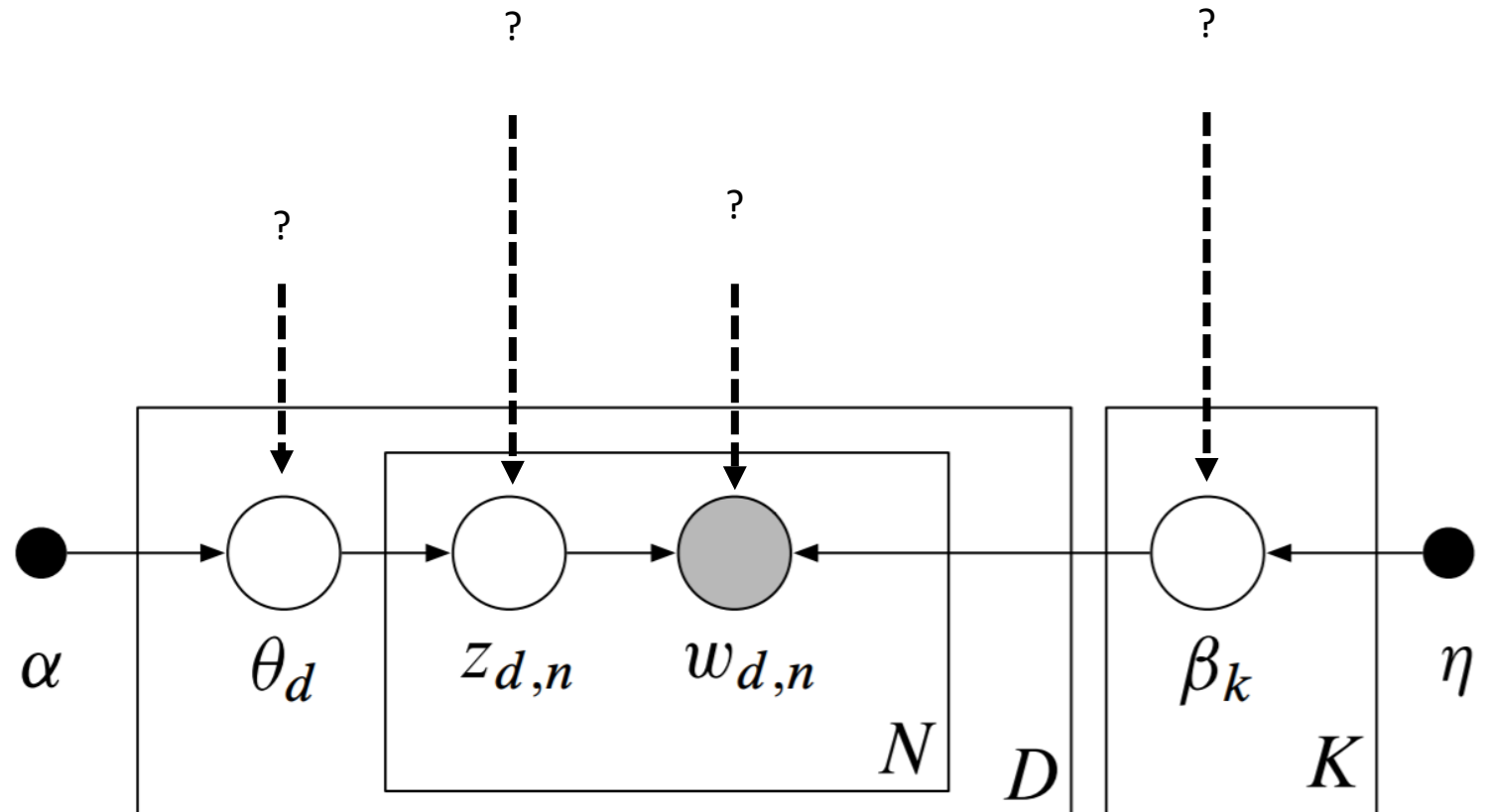
Archaeogenomic analysis of scarlet macaw bones demonstrates that the genetic diversity of these birds acquired by people in the southwestern United States (SW) between 900 and 1200 CE was exceedingly low. Only one mitochondrial DNA haplogroup (Haplo6) is present of the five historically known haplogroups in the lowland forests of Mexico and Central America. Phylogenetic analyses indicate the ancient macaw lineage in the SW shared genetic affinities with this wild lineage. These data support the hypothesis that a translocated

# Latent Dirichlet allocation

- How do we model topic proportions in a document corpus?
- What is observed?
- What is unobserved?

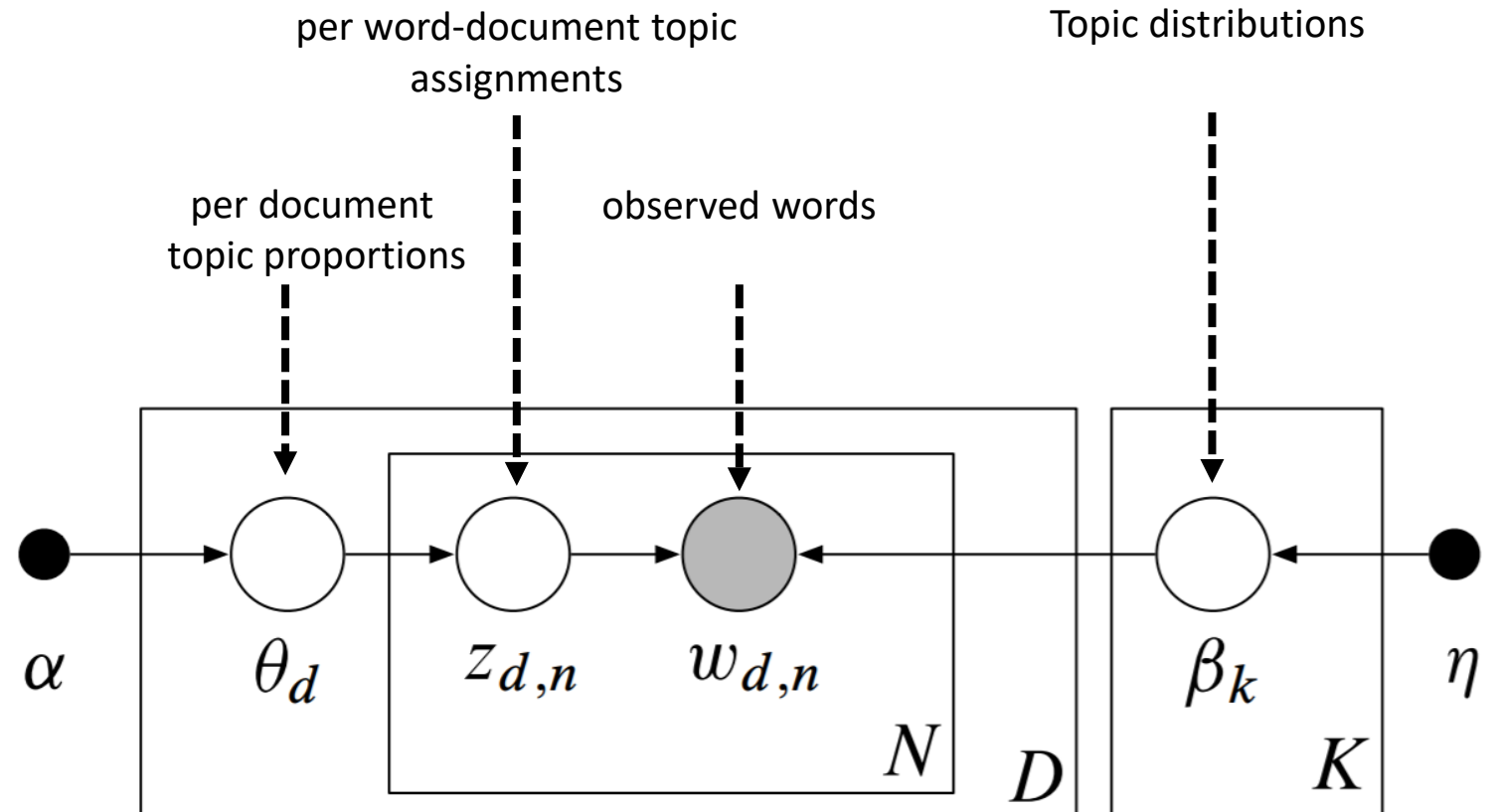
# Latent Dirichlet allocation

- How do we model topic proportions in a document corpus?
- What is observed?
- What is unobserved?



# Latent Dirichlet allocation

- How do we model topic proportions in a document corpus?
- What is observed?
- What is unobserved?

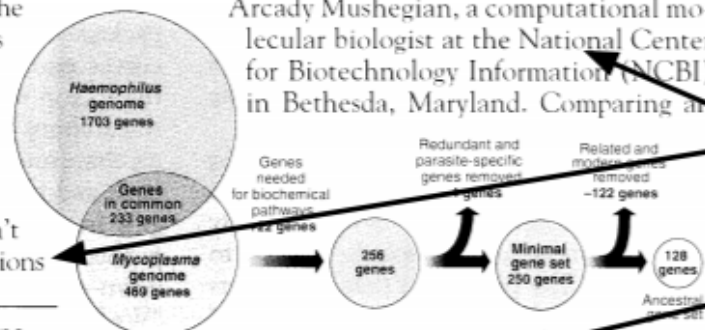


## Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many genes does an organism need to survive? Last week at the genome meeting here,\* two genome researchers with radically different approaches presented complementary views of the basic genes needed for life. One research team, using computer analyses to compare known genomes, concluded that today's organisms can be sustained with just 250 genes, and that the earliest life forms required a mere 128 genes. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those predictions

"are not all that far apart," especially in comparison to the 75,000 genes in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a genetic numbers game, particularly as more and more genomes are completely mapped and sequenced. "It may be a way of organizing any newly sequenced genome," explains Arcady Mushegian, a computational molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an



\* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

**Stripping down.** Computer analysis yields an estimate of the minimum modern and ancient genomes.

SCIENCE • VOL. 272 • 24 MAY 1996

## Topics

gene 0.04  
dna 0.02  
genetic 0.01  
...

life 0.02  
evolve 0.01  
organism 0.01  
...

brain 0.04  
neuron 0.02  
nerve 0.01  
...

data 0.02  
number 0.02  
computer 0.01  
...

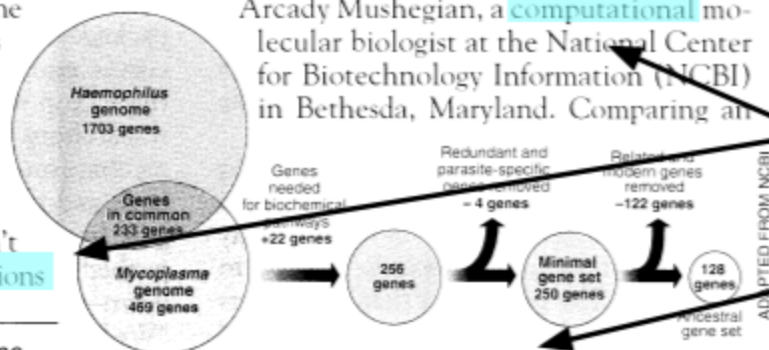
## Documents

### Seeking Life's Bare (Genetic) Necessities

COLD SPRING HARBOR, NEW YORK—How many **genes** does an **organism** need to **survive**? Last week at the genome meeting here,\* two genome researchers with radically different approaches presented complementary views of the basic genes needed for **life**. One research team, using **computer** analyses to compare known **genomes**, concluded that today's **organisms** can be sustained with just 250 genes, and that the earliest life forms required a mere 128 **genes**. The other researcher mapped genes in a simple parasite and estimated that for this organism, 800 genes are plenty to do the job—but that anything short of 100 wouldn't be enough.

Although the numbers don't match precisely, those **predictions**

"are not all that far apart," especially in comparison to the 75,000 **genes** in the human genome, notes Siv Andersson of Uppsala University in Sweden, who arrived at the 800 number. But coming up with a consensus answer may be more than just a **genetic numbers** game, particularly as more and more **genomes** are completely mapped and sequenced. "It may be a way of organizing any newly **sequenced genome**," explains Arcady Mushegian, a **computational** molecular biologist at the National Center for Biotechnology Information (NCBI) in Bethesda, Maryland. Comparing an

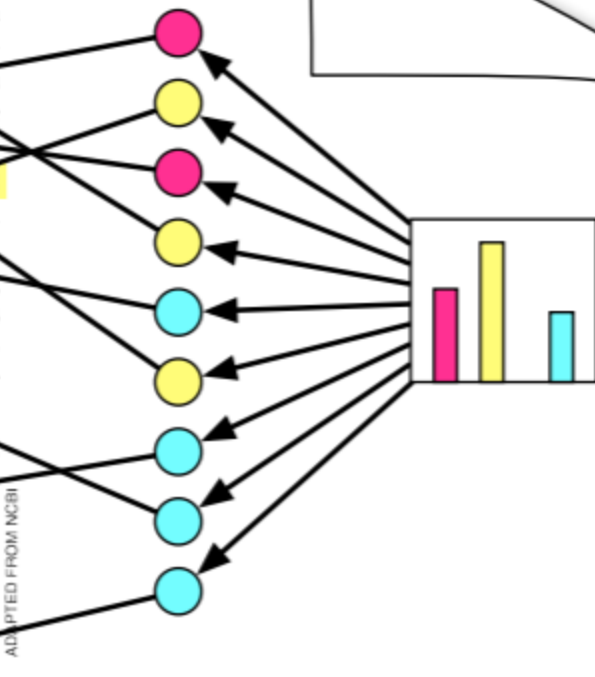


\* Genome Mapping and Sequencing, Cold Spring Harbor, New York, May 8 to 12.

**Stripping down.** **Computer analysis** yields an estimate of the minimum modern and ancient genomes.

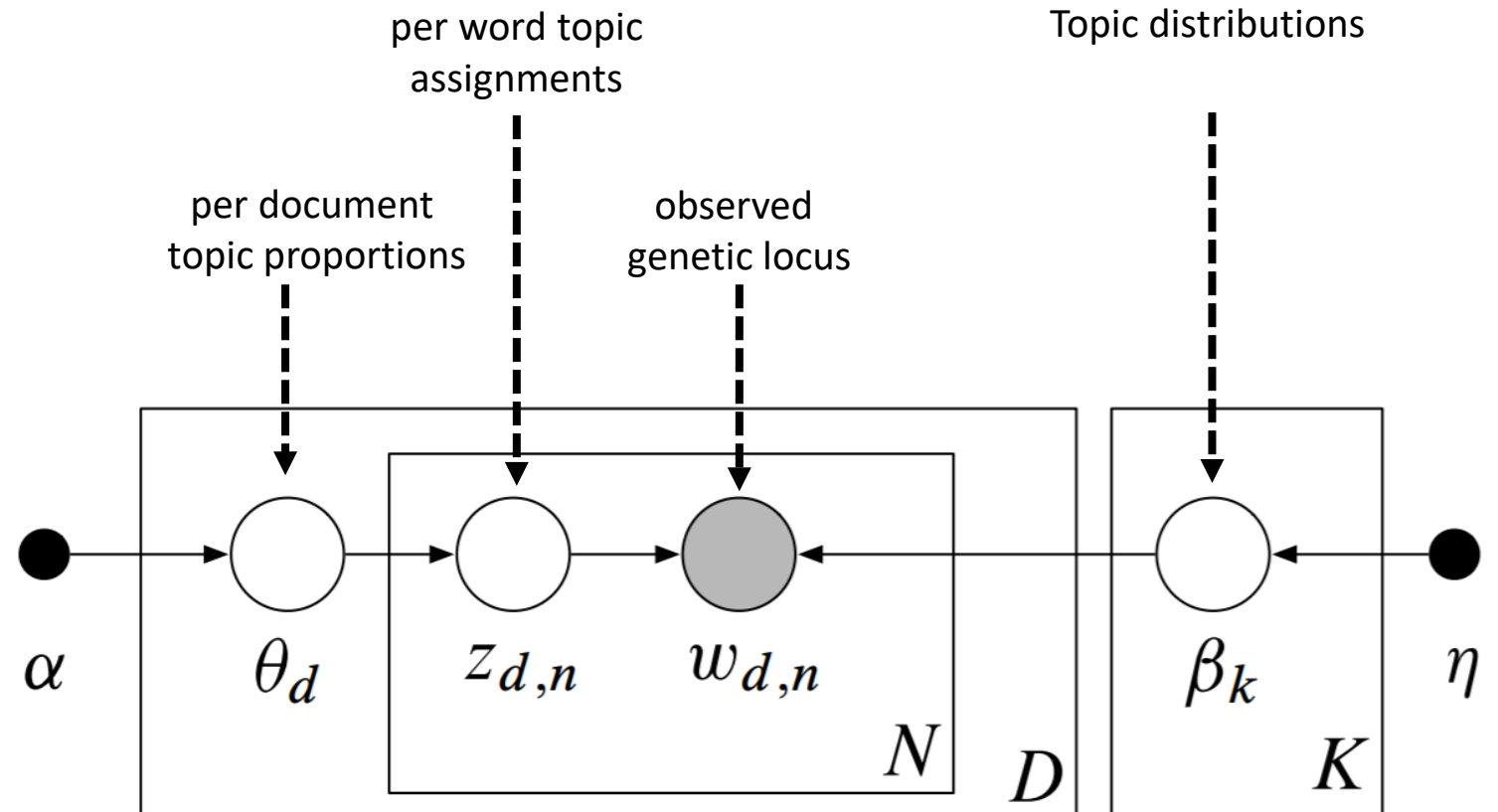
SCIENCE • VOL. 272 • 24 MAY 1996

## Topic proportions and assignments



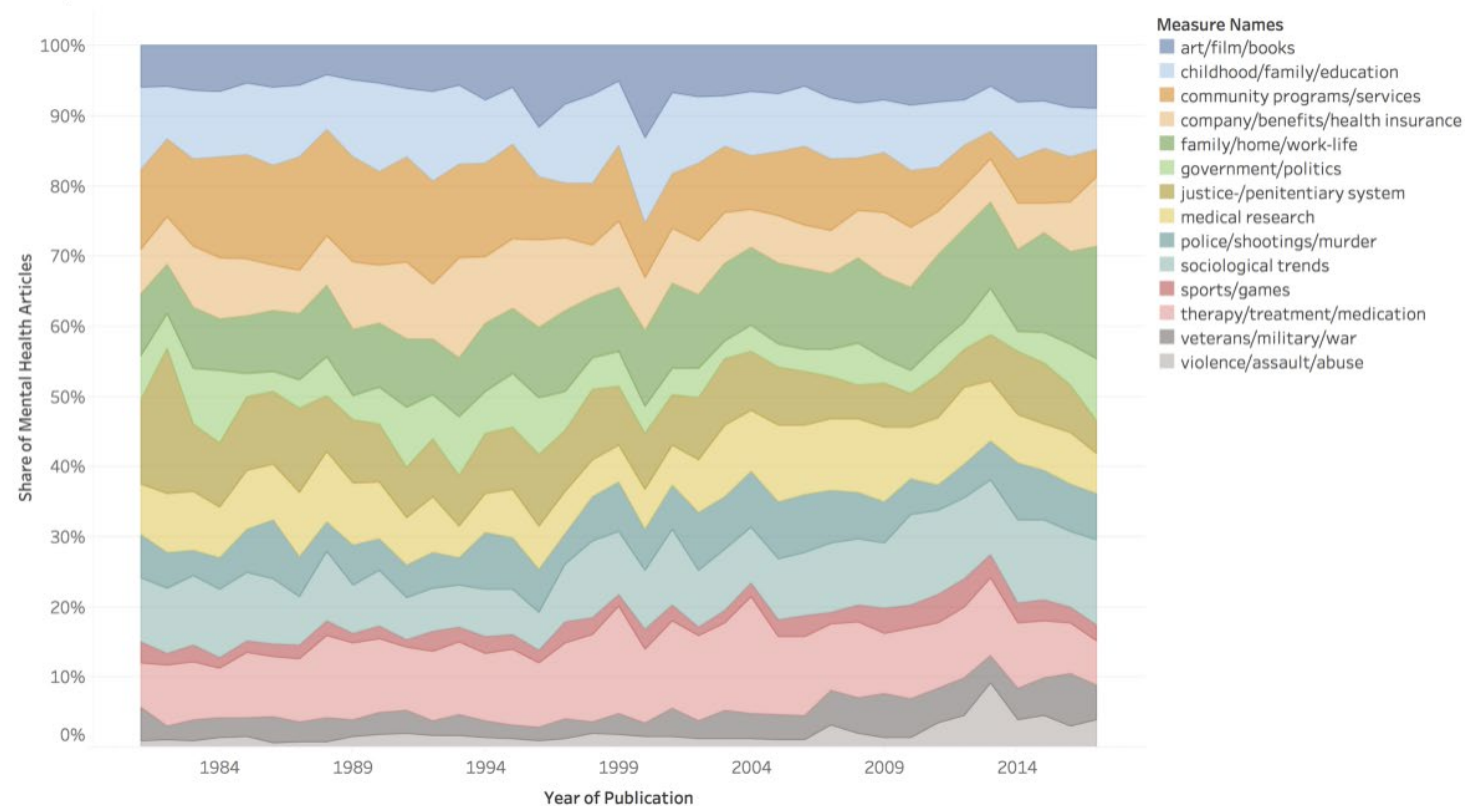
# Latent Dirichlet allocation

- What are some of the quantities useful for interpretation of a set of documents?
  - $\theta$  gives the mixture of topics for each document
  - $z$  gives local document information for a word
  - $\beta$  gives the topic distribution over the words in a corpus



# NYT, mental health articles over time

Topic Shares of Mental Health Articles over Time

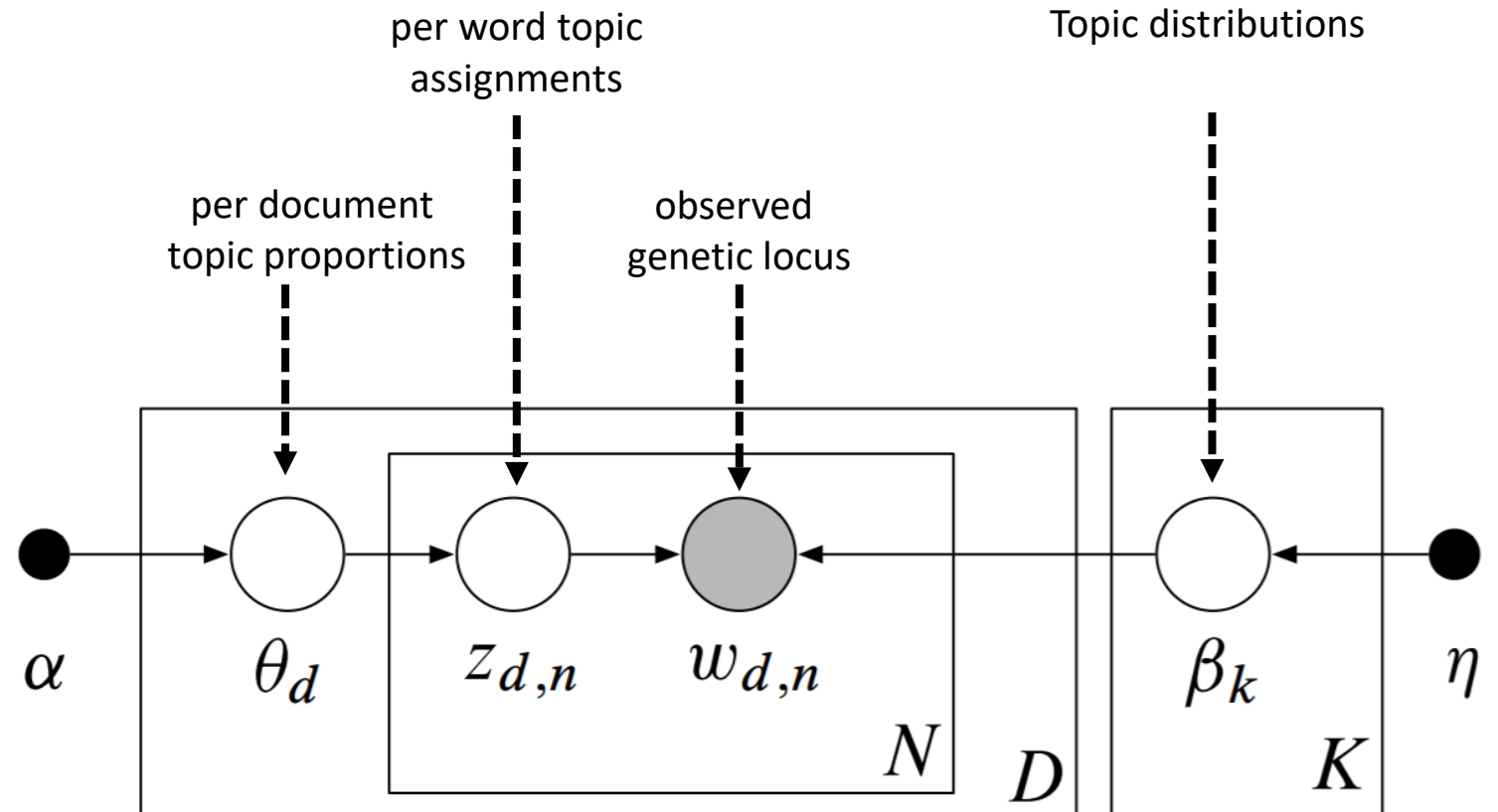


# Topic co-occurrence



# Latent Dirichlet allocation

- What are some of the quantities useful for interpretation of a set of documents?
  - $\theta$  gives the mixture of topics for each document
  - $z$  gives local document information for a word
  - **$\beta$  gives the topic distribution over the words in a corpus**



# NYT topics



# Why Bayesian ML?

# Why Bayesian ML?

- Probabilistic predictions

# Probabilistic Predictions

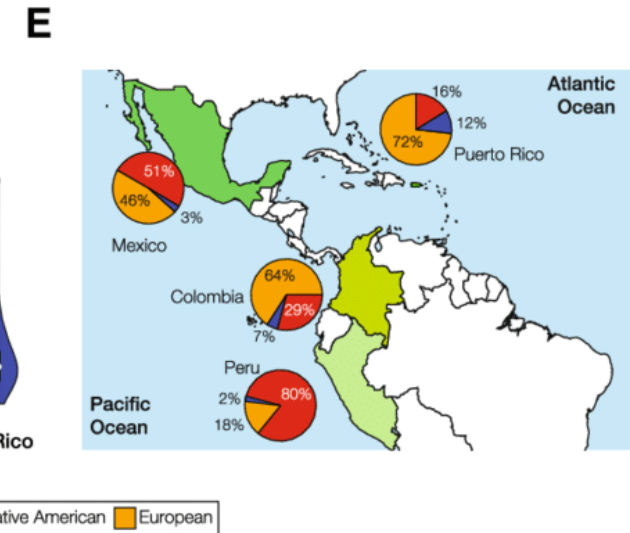
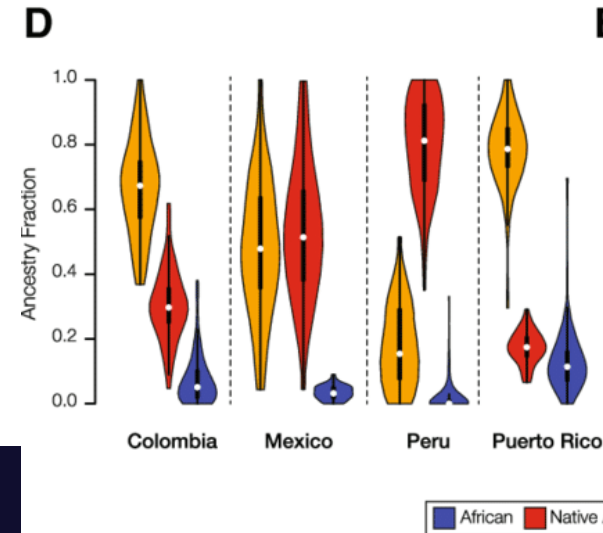
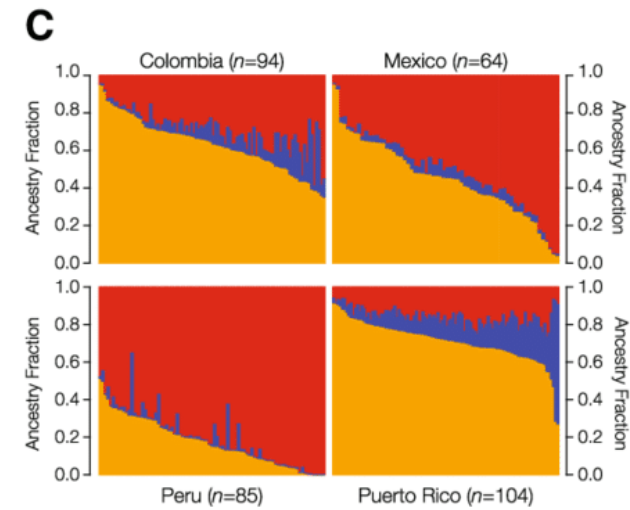
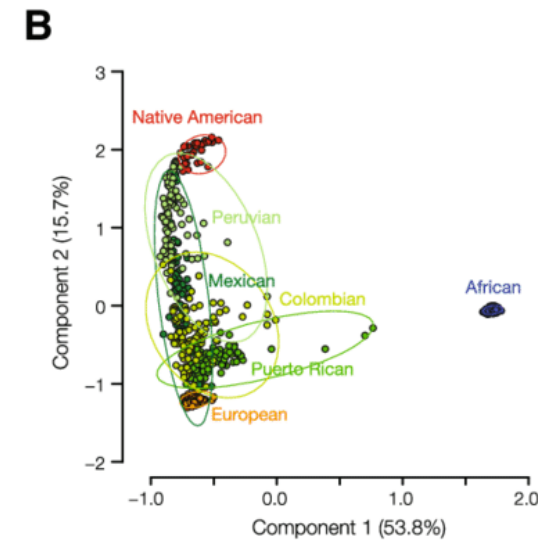
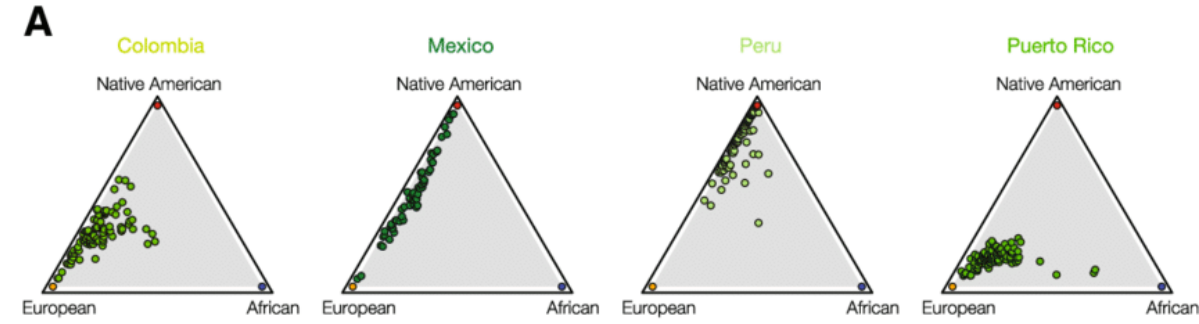
- Uncertainty around estimate
  - From a formal probabilistic model
- When is this useful?

# Probabilistic Predictions

- Uncertainty around estimate
  - From a formal probabilistic model
- When is this useful?
  - Interpreting results
    - E.g. populations which have samples that are commonly mixed, can be said to be 'admixed'

# Probabilistic Predictions

- Uncertainty around estimate
  - From a formal probabilistic model
- When is this useful?
  - Interpreting results
    - E.g. populations which have samples that are commonly mixed, can be said to be 'admixed'



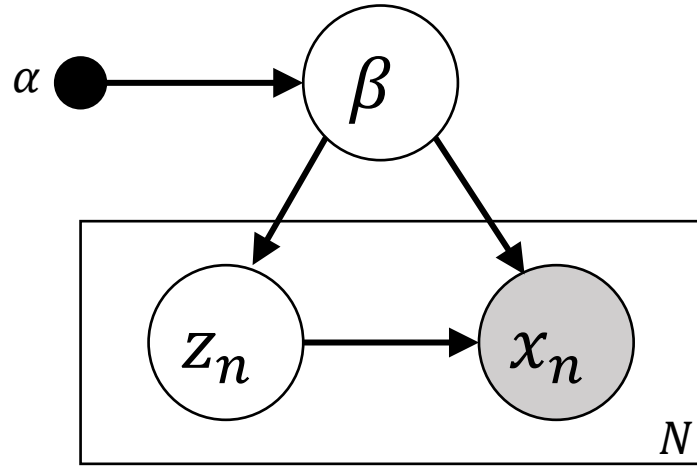
# Probabilistic Predictions

- Uncertainty around estimate
  - From a formal probabilistic model
- When is this useful?
  - Interpreting results
    - E.g. populations which have samples that are commonly mixed, can be said to be 'admixed'
  - Decision making
    - "the probable is the hypothesis on which it is rational for us to act" Keynes 1921
    - Few decisions can be made based on 'logic' alone. Instead they must incorporate probabilities and uncertainty
    - E.g. when to allow bail given uncertainty about flight risk or danger to others

# Why Bayesian ML?

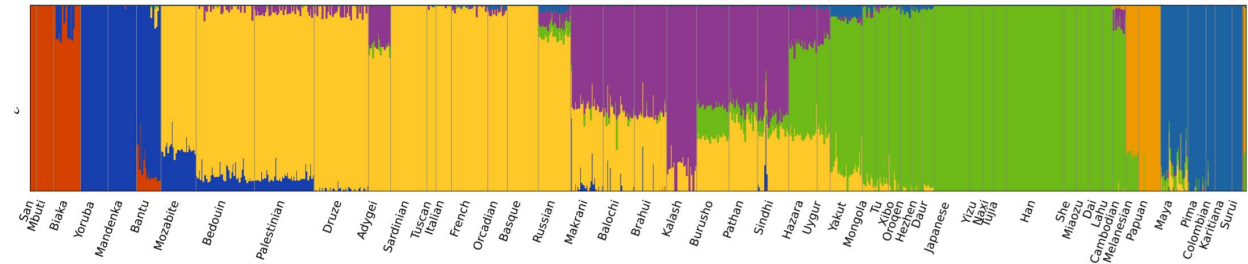
- Probabilistic predictions
- It's becoming easier to iterate on model design

# Decouple representation from inference

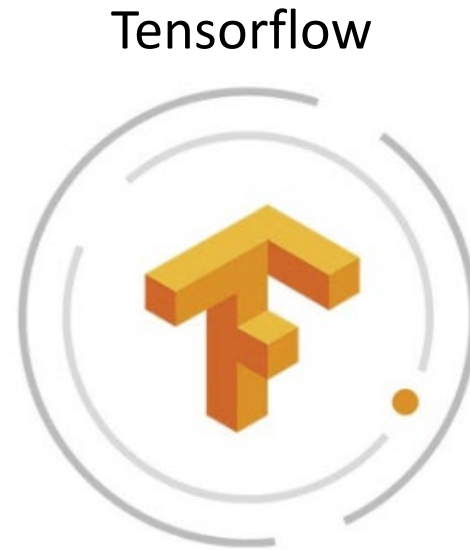


$$p(x, z, \beta) = p(z, \beta | x)p(x)$$

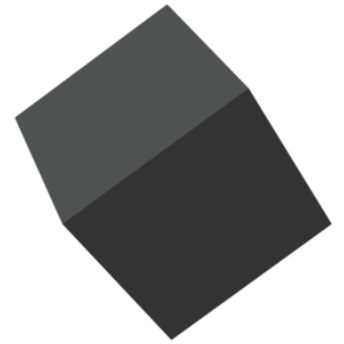
$$p(z, \beta | x) = \frac{p(x, z, \beta)}{p(x)}$$



# Implementation



Edward



Pyro



# Edward model for...?

$$p(\mathbf{w}) = \text{Normal}(\mathbf{w} \mid \mathbf{0}, \sigma_w^2 \mathbf{I}),$$

$$p(b) = \text{Normal}(b \mid 0, \sigma_b^2),$$

$$p(\mathbf{y} \mid \mathbf{w}, b, \mathbf{X}) = \prod_{n=1}^N \text{Normal}(y_n \mid \mathbf{x}_n^\top \mathbf{w} + b, \sigma_y^2).$$

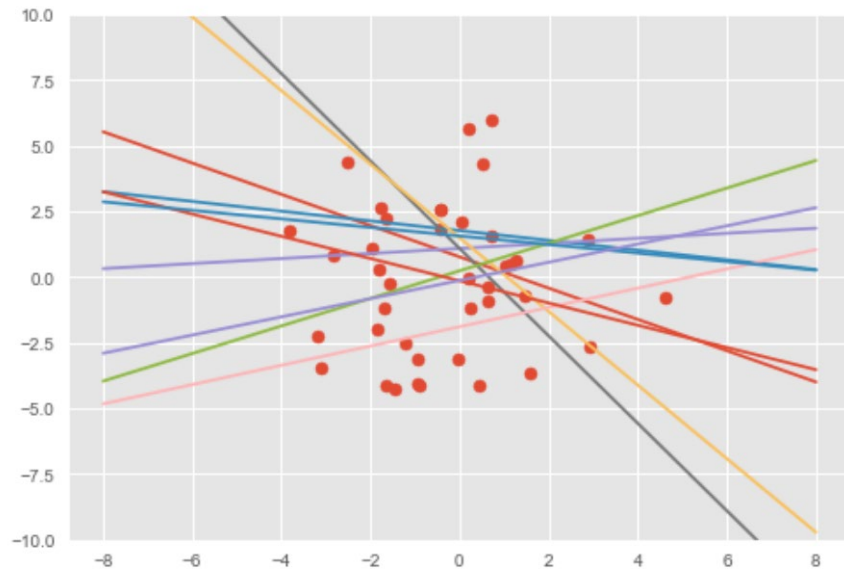
```
1 from edward.models import Normal
2
3 X = tf.placeholder(tf.float32, [N, D])
4 w = Normal(loc=tf.zeros(D), scale=tf.ones(D))
5 b = Normal(loc=tf.zeros(1), scale=tf.ones(1))
6 y = Normal(loc=ed.dot(X, w) + b, scale=tf.ones(N))
```

# Edward model for Bayesian linear regression

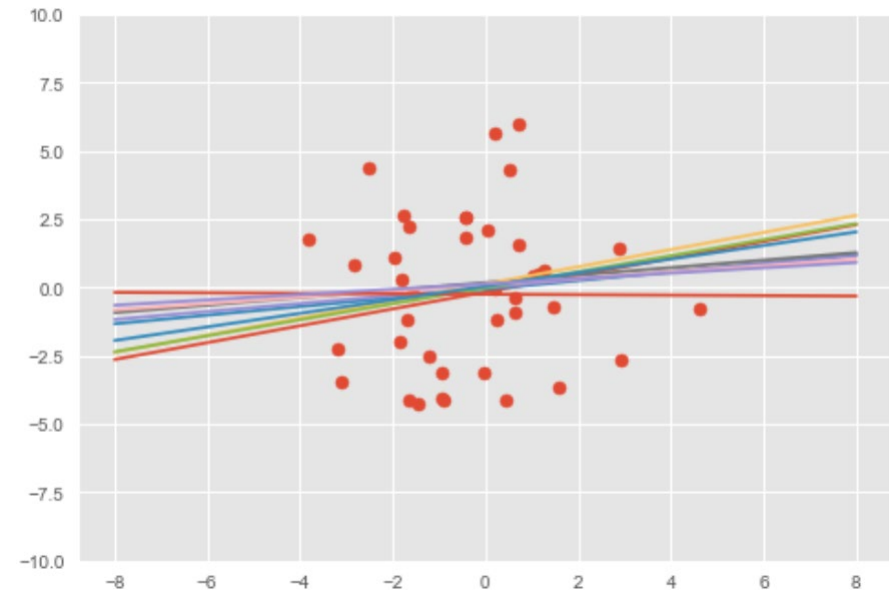
$$p(\mathbf{w}) = \text{Normal}(\mathbf{w} \mid \mathbf{0}, \sigma_w^2 \mathbf{I}),$$
$$p(b) = \text{Normal}(b \mid 0, \sigma_b^2),$$
$$p(\mathbf{y} \mid \mathbf{w}, b, \mathbf{X}) = \prod_{n=1}^N \text{Normal}(y_n \mid \mathbf{x}_n^\top \mathbf{w} + b, \sigma_y^2).$$

```
1 from edward.models import Normal
2
3 X = tf.placeholder(tf.float32, [N, D])
4 w = Normal(loc=tf.zeros(D), scale=tf.ones(D))
5 b = Normal(loc=tf.zeros(1), scale=tf.ones(1))
6 y = Normal(loc=ed.dot(X, w) + b, scale=tf.ones(N))
```

Samples from prior



Samples from posterior



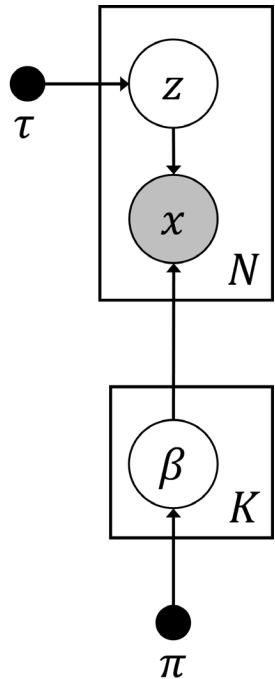
# Why Bayesian ML?

- Probabilistic predictions
- It's becoming easier to iterate on model design
- Modularity

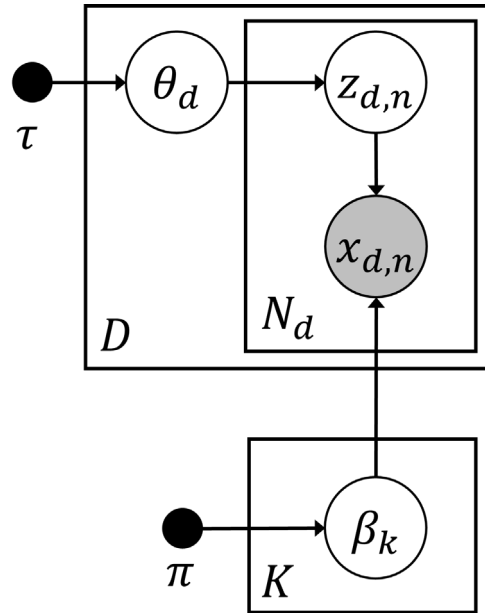
# Modularity

- Simpler models can be used as components for more sophisticated models

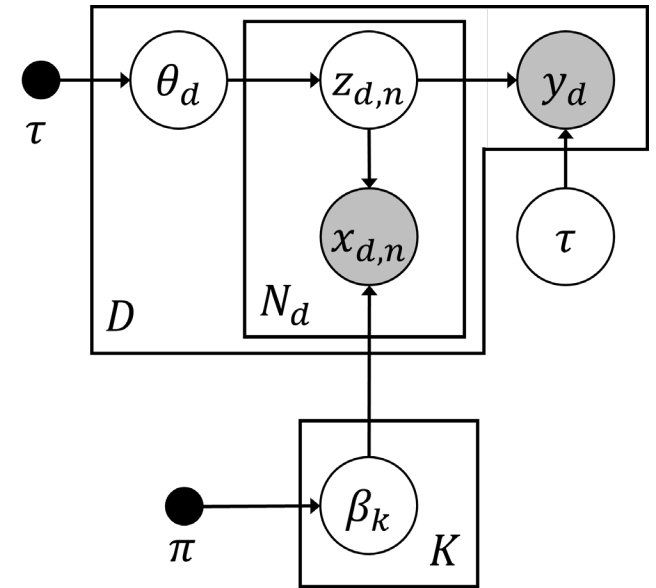
Basic mixture model



LDA

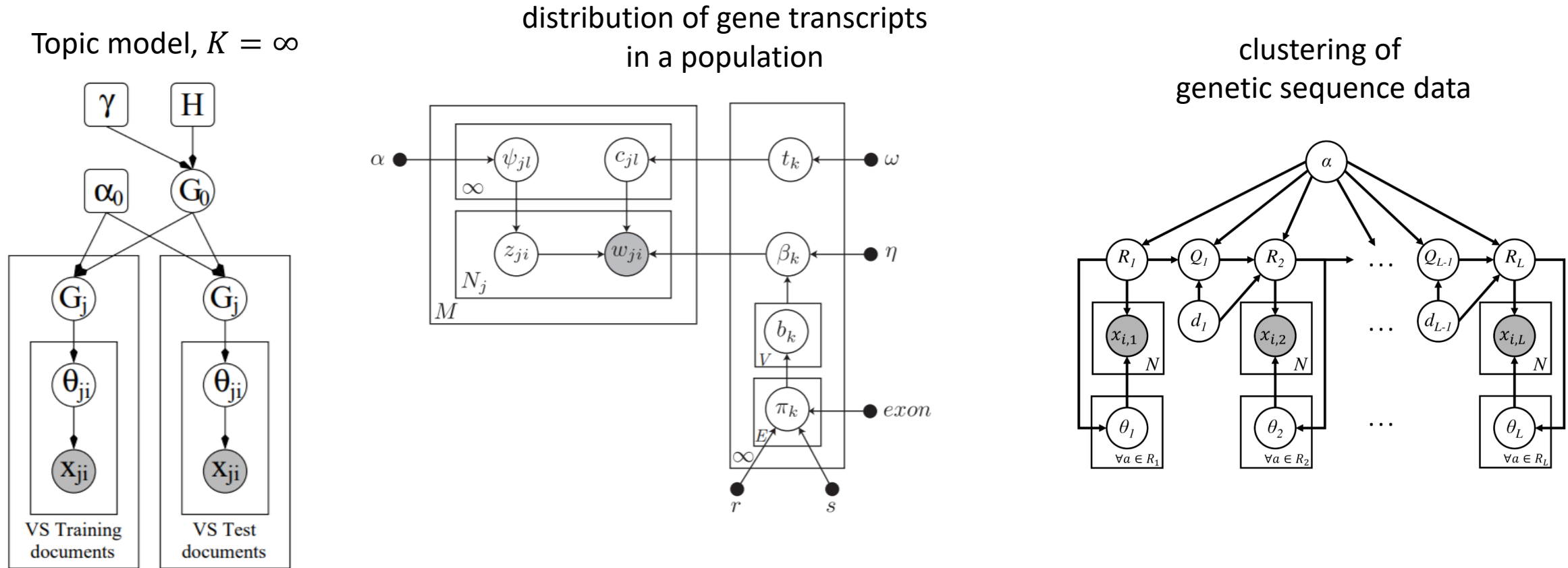


Supervised LDA



# Modularity

- And even more sophisticated



# Why Bayesian ML?

- Probabilistic predictions
- It's becoming easier to iterate on model design
- Modularity
- Regular workflow for interdisciplinary work (domain experts and computational modelers)

# Why Bayesian ML?

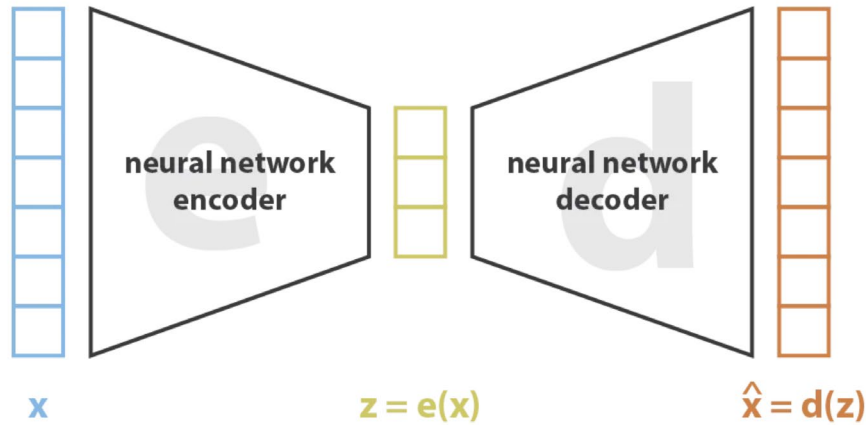
- Probabilistic predictions
- It's becoming easier to iterate on model design
- Modularity
- Regular workflow for interdisciplinary work (domain experts and computational modelers)
- Clear language for defining model, i.e., what is hidden, what is known, how they relate to each other
  - Related: explainability and interpretability

# Why Bayesian ML?

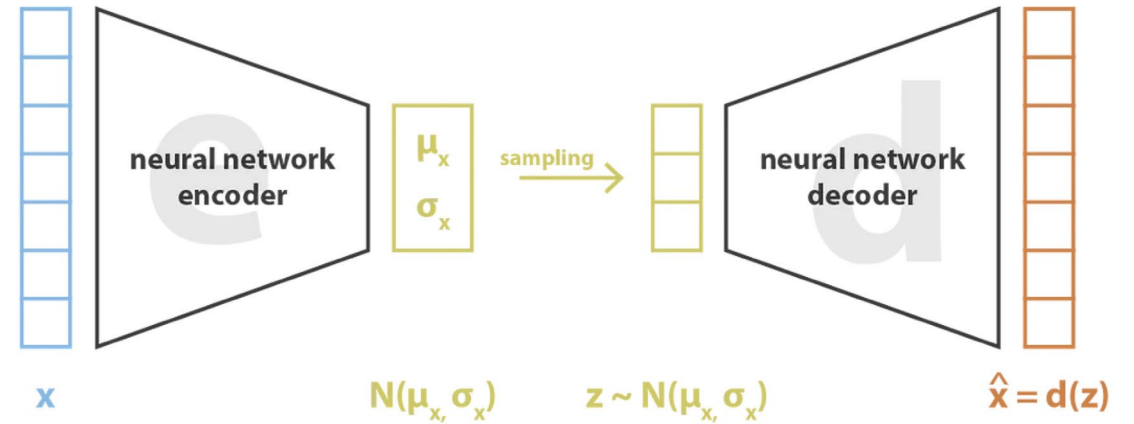
- Probabilistic predictions
- It's becoming easier to iterate on model design
- Modularity
- Regular workflow for interdisciplinary work (domain experts and computational modelers)
- Clear language for defining model, i.e., what is hidden, what is known, how they relate to each other
  - Related: explainability and interpretability
- Procedure for criticizing models

# Reasons why you may not use Bayesian ML

- Features are poorly understood
  - What is the prior for an image? For text?



$$\text{loss} = \|x - \hat{x}\|^2 = \|x - d(z)\|^2 = \|x - d(e(x))\|^2$$



$$\text{loss} = \|x - \hat{x}\|^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)] = \|x - d(z)\|^2 + \text{KL}[N(\mu_x, \sigma_x), N(0, I)]$$

# Reasons why you may not use Bayesian ML

- Features are poorly understood
  - What is the prior for an image? For text?
- A lot of data exists for problem
  - Likelihood dominates posterior, inference typically more difficult
- Final comment: many models include both probabilistic (Bayesian/Frequentist) and non probabilistic (e.g. neural network) components

# Topics

- Graphical models
  - Bayesian and Markov networks
  - Independence in graphs and d-separation
  - Statistical concepts and models
- Inference
  - Markov chain Monte Carlo
  - Variational inference
  - Stochastic variational inference
  - Maximization expectation and/or small variance asymptotics
  - (maybe) Coresets

# Topics

- Modelling
  - Topic and genetic mixture models
  - Multilevel Regression
  - ...
- Posterior checking
  - Posterior predictive checks
  - Predictive sample re-use (cross validation)
- Advanced topics
  - Variational autoencoders
  - Bayesian nonparametrics
  - (probably) Latent Diffusion
  - (maybe) Normalizing flows

# Questions?

